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Constraints on the Ediacaran inertial interchange true polar wander hypothesis: A new paleomagnetic study in Morocco (West African Craton)

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ABSTRACT

We conducted a paleomagnetic study on late Ediacaran and early Cambrian volcanic deposits and lava flows from the Ouarzazate and Taroudannt groups in the Anti-Atlas, Morocco. Four stable magnetic mean directions have been isolated using thermal demagnetization technique. A first component "A" is most probably a remagnetization acquired during the Hercynian related tectonic phases, and yields a paleomagnetic pole ($\lambda = -29.3°$ N, $\phi = 56.6°$ E, $A_{95} = 4.1°$) indistinguishable from the 330–300 Ma segment of the Gondwana apparent polar wander path (APWP). The Cambrian Djebel Boho formation of the Taroudannt group and the Ediacaran Tadoughast and Fajjoud formations of the Ouarzazate group yield two poles of similar directions respectively "B1" ($\lambda = 21.9°$N, $\phi = 31.0°$E, $A_{95} = 15.6°$) and "B2" ($\lambda = 27.3°$N, $\phi = 27.1°$E, $A_{95} = 14.9°$). In the Adrar-n-Takoucht formation (oldest part of the Ouarzazate group), another component is observed, yielding a paleomagnetic pole "C" ($\lambda = -57.6°$N, $\phi = 295.6°$E, $A_{95} = 15.7°$). The primary nature of these last three magnetizations is supported by paleomagnetic tests and we interpreted them as magnetizations acquired during or shortly after the deposit of volcaniclastics or emplacement of lava flows. Using these new data, we reappraised the West African Craton (WAC) APWP and found two large paleomagnetic shifts of some 90°, the first one between 615 and 571 Ma and the second one between 571 and 565 Ma. A comparison with already published paleomagnetic data from Laurentia, Baltica and Gondwana, along with a discussion on their paleogeographic relationships, showed that our WAC APWP loop is likely recorded in the APWP's of these blocks. The superimposition of these APWP's leads to a geologically consistent paleogeography which may be explained by two rapid inertial interchange true polar wander even if we cannot exclude that magnetic field perturbations may have also occurred during the Ediacaran.

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1. Introduction

The Ediacaran (635–542 Ma) is a geological period of dramatic changes. Many phenomenon occurred, involving the cryosphere, the biosphere and the geosphere, including one or more large-scale glaciations (e.g. Hoffman and Li, 2009), the explosion of the Ediacaran fauna (e.g. Knoll, 1992), the final dislocation of the supercontinent Rodinia (e.g. Cawood and Pisarevsky, 2006) and the assembly of Gondwana (e.g. Meert, 2003; Cordani et al., 2013).

For this geological period, only a few reliable and well-dated paleomagnetic data are documented in the literature, leading to controversial paleogeographic reconstructions. One of the most documented (and also problematic) datasets comes from Laurentia which consists of two groups of paleomagnetic poles located at high and low latitudes separated by around 90°. If all these poles are considered primary and under the central assumption of a geocentric axial dipole (GAD), multiple migrations of the continent Laurentia from the high to the low latitudes should have occurred implying extremely high plate velocities in excess of 60 cm/yr (Abrajevitch and Van der Voo, 2010). Large oscillations are also proposed for the APWP of Baltica (Abrajevitch and Van der Voo, 2010). Several explanations have been proposed in the literature such as: (1) the presence of unreliable paleomagnetic data and/or inaccurate age datings used to propose continental reconstructions (e.g. Hodych et al., 2004), (2) the oscillation between an Earth's

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magnetic field dominated by an axial and an equatorial dipole (Abrajevitch and Van der Voo, 2010) (3) recording of an equatorial dipole field as a transitional field between opposite polarity states (Halls et al., 2015) (4) very rapid true polar wander (TPW) episodes involving the tumbling of the bulk Earth with respect to its spin axis (e.g. Evans, 2003). If one of these latter mechanisms occurred, the rapid and large oscillations observed for Laurentia and Baltica should be recorded in any other rocks of the same age from other continents.

In this study, we tested these hypotheses by conducting a new paleomagnetic study on Ediacaran and Cambrian volcanic series from an independent continent, the West African Craton (WAC). The studied region is situated in the Anti-Atlas mountain range (Morocco) where recent geological mapping was conducted by the French Geological Survey (BRGM: Bureau de Recherches Géologiques et Minières) bringing new geological and geochronological constraints on volcanic series of similar age as those giving discordant paleomagnetic results in Laurentia and Baltica.

The timing of accretion of the WAC to the Gondwana is still debatted. Collision ages generally converge around the early to mid Ediacaran, around 600 Ma (e.g. Caby, 2003; Cordani et al., 2013). A much younger age of accretion is also proposed, invoking the existence of a large ocean between the WAC and Amazonia, and the other blocks of the Gondwana until early Cambrian times (Tohver et al., 2006; Trindade et al., 2006; Tohver et al., 2012). Nevertheless, this last estimation is based on a very low amount of good quality paleomagnetic data from the WAC and Amazonia and is still controversial. In this study, we also discuss the age of accretion of the WAC to the Gondwana thanks to our new paleomagnetic dataset.

2. Geological context and sampling

The Moroccan Anti-Atlas marks the northern boundary of the Eburnean West African craton, and has been essentially impacted by the Pan-African orogeny (Fig. 1a and b, e.g. Villeneuve and Cornée, 1994; Inglis et al., 2004; Gasquet et al., 2005, 2008; Nance et al., 2008; El Hadi et al., 2010). The Anti-Atlas is characterized by outcrops of Proterozoic basement unconformably overlain by late Ediacaran to Cambrian volcanic and sedimentary rocks (Fig. 1b). The Pan-African geodynamic evolution of this border is marked (as defined by Thomas et al., 2004; Samson et al., 2004; Gasquet et al., 2005, 2008; D'Lemos et al., 2006; Blein et al., 2014a; Triantafyllou et al., 2015) by: (1) Tonian (?)-lower Cryogenian (1000–800 Ma) rifting of the WAC and formation of a passive margin platform, (2) development of oceanic island-arcs (770–710 Ma), (3) ocean closure marked by polyphase accretion of oceanic rocks (710–630 Ma), (4) development of an active margin along the amalgamated WAC (630–580 Ma), (5) a post-subduction period after a ridge-trench collision characterized by a magmatic event attributed to asthenospheric rise related to the development of a slab-window (580–545 Ma). During this last episode, a large volume of volcanic rocks is produced corresponding to the Ouarzazate group. During the subsidence of the Anti-Atlas, carbonate-related rocks covered these Precambrian rocks (e.g. Thomas et al., 2002). The Anti-Atlas is slightly folded during the Hercynian orogeny (e.g. Michard et al., 2008) and again during the opening of the Atlantic ocean and the Alpine cycle.

The Ouarzazate group consists of volcanic and sedimentary deposits. It outcrops with a relatively low thickness (0–2000 m) in many Precambrian inliers of the Anti-Atlas (Choubert, 1947, 1964; Thomas et al., 2002). Recent mapping conducted by the BRGM allows the acquisition of a large volume of geological, geochemical and geochronological data in the Anti-Atlas and especially in the Agadir Melloul, Iguerda, Sirwa and Bou Azzer inliers (Blein et al., 2014a,b). Four formations were distinguished in mapping volcanic and sedimentary deposits of the Ouarzazate Group in Iguerda, Agadir Melloul and south Sirwa inliers (Fig. 1c): Adrar-n-Takoucht, Anammar, Tadoughast and Fajjoud.

The Adrar-n-Takoucht formation is composed of pyroclastic volcanic deposits which are exposed in the Sirwa, Agadir Melloul and Idikal inliers (Fig. 1c). In the Sirwa inlier, this formation consists of three distinct pyroclastic sequences characterized by massive basal pyroclastic flows overlain by lithic tuffs, crystal tuffs, hyaline tuffs and well stratified cinerites. In the Idikal inlier, this formation overlain directly the Paleoproterozoic basement. Blein et al. (2014b) obtained a SHRIMP U-Pb age on zircon of 572 ± 5 Ma at the bottom and 570 ± 6 Ma close to the top of the formation. Due to the few amount of U-Pb ages in this formation localized at different stratigraphic positions, we think that the calculation of an arithmetic average and a standard deviation to characterize the age of the formation is not statistically appropriated. We preferred to take into account the full range of uncertainties on the U-Pb ages to estimate the age of the formation, giving an age comprised between 577 and 564 Ma. This formation is unconformably overlain by the Tadoughast Formation in the Sirwa inlier, and by the Anammar Formation in the Agadir Melloul inlier.

The Tadoughast formation consists of volcano-sedimentary breccias, pyroclastic breccias, pyroclastic tuffs with lithic and/or crystal clasts, crystal tuffs, cinerites, rhyolitic domes and breccias. This formation sometimes overlains directly the Paleoproterozoic basement like in the Agadir Melloul inlier (Blein et al., 2014b). Pyroclastic deposits are more massive towards the top of this formation, marking deposits from massive pyroclastic flows (Fig. 2). Five SHRIMP U-Pb ages on zircon have been obtained for the Tadoughast formation (Blein et al., 2014b). Taking the full range of uncertainties on the U-Pb datings, an age comprised between 572 and 558 Ma was calculated.

The Fajjoud formation overlies the Tadoughast Formation or the Paleoproterozoic basement. This formation is composed of pyroclastic sequences with massive lithic breccia, lithic-rich or crystal rich pyroclastic tuffs, ash-flow and ash-fall tuffs. This formation lays on the Tadoughast formation and sometimes overlains directly the Paleoproterozoic basement like in the Agadir Melloul inlier (Blein et al., 2014b). As in the Tadoughast formation, the top of the Fajjoud Formation is characterized by massive pyroclastic flows (Fig. 2). A SHRIMP U-Pb age on zircon of 556 ± 5 Ma is obtained on a highly welded tuff located close to the top (Blein et al., 2014b), constraining the maximum age for the Fajjoud formation.

The Taroudannt group is composed of the carbonate rocks of the Adoudou formation and the mixed terrigenous-carbonate rocks of the Taliwine formation. This sedimentary late Ediacaran–early Cambrian sequence unconformably overlies the volcano-sedimentary series of the Ouarzazate group. In the Bou Azzer inlier (Fig. 1b and d), pyroclastic tuffs and volcanic flows of trachytic and basaltic composition are also interbedded with the Adoudou strata corresponding to the volcanic rocks of the Djebel Boho formation (Fig. 2; Choubert and Faure-Muret, 1970; Blein et al., 2014a). Three U-Pb ages on zircon have been obtained for this formation (529 ± 3 Ma from Ducrot and Lancelot, 1977, recalculated by Chèvremont et al., 2013; 531 ± 5 Ma from Gasquet et al., 2005; 541 ± 6 Ma from Blein et al., 2014a). Taking the full range of uncertainties on the U-Pb datings, an age comprised between 547 and 526 Ma is calculated.

We collected oriented samples with a portable drill at 103 sites distributed in the volcanic rocks of the Adrar-n-Takoucht, Tadoughast, Fajjoud and Djebel Boho formations (Figs. 1c, d and 2). Each site corresponds to a single pyroclastic deposit or lava flow.

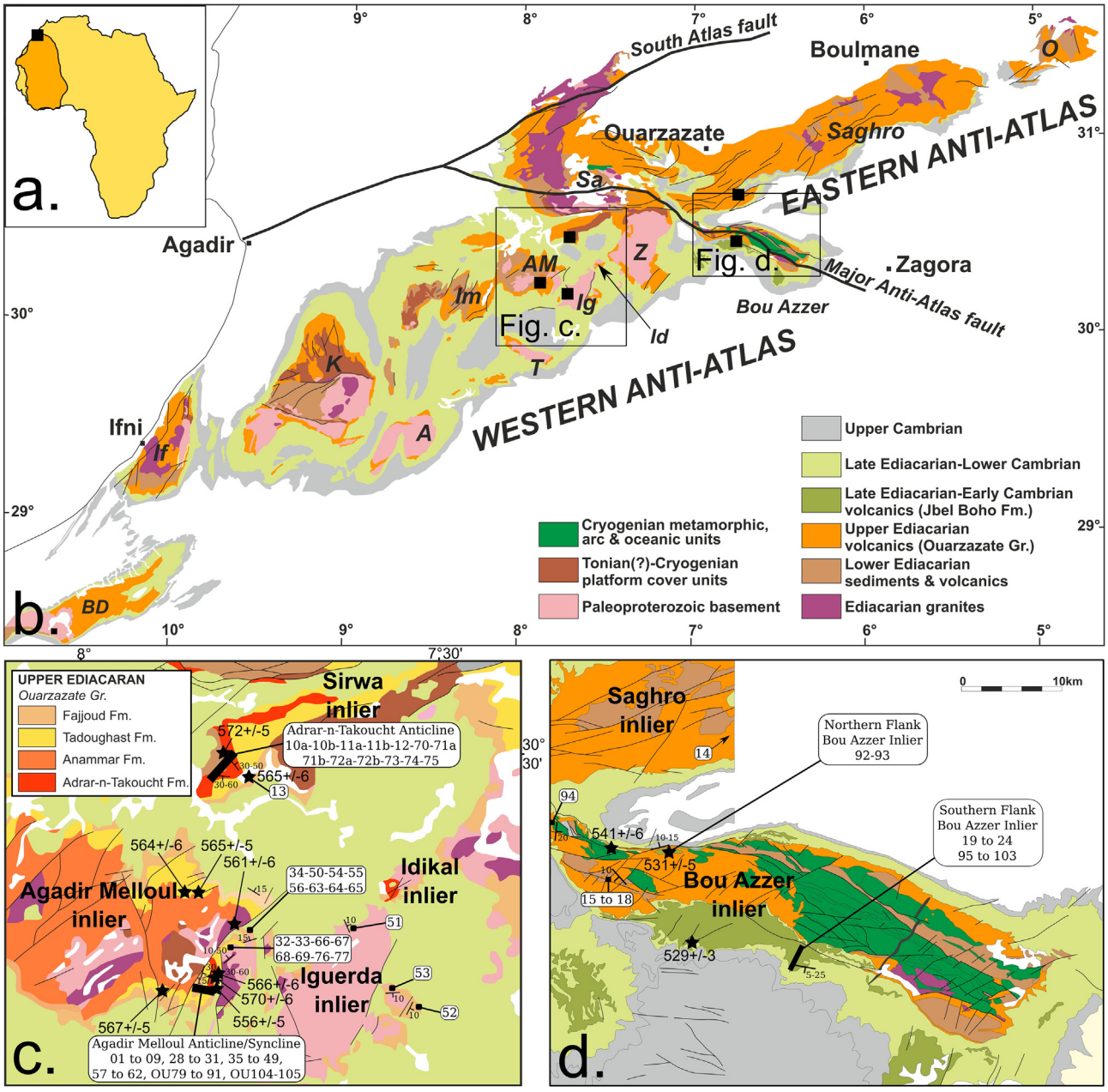


Fig. 1. a. Localization of the West African craton (orange) in Africa (yellow). b. Geological map of the Anti-Atlas. Modified from Hollard et al. (1985), Walsh et al. (2002), Gasquet et al. (2008) and Blein et al. (2014b). Areas delimited by the black rectangles are zoomed in the figures c. and d. The black squares correspond to the sampled localities. Abbreviations mean: BD: Bas Drâa; If: Ifni; K: Kerdous; A: Tagragra d'Akka; Im: Igherm; T: Tagragra de Tata; Ig: Iguerda; Id: Idikal; AM: Agadir-Melloul; Sa: Sirwa; O: Ougnat. c. Zoom on the Agadir Melloul, Iguerda, Idikal and Sirwa inliers. Modified from Blein et al. (2014b). The name of the sampled sites from each sampled locality is detailed in white boxes. d. Zoom on the Bou Azzer inlier. Modified from Blein et al. (2014a). Black stars show the localization of the U-Pb datings on zircon carried out by Blein et al. (2014b) in c. and by Blein et al. (2014a) (541 ± 6 Ma), Thomas et al. (2002) (531 ± 5 Ma) and Ducrot and Lancelot (1977) recalculated by Chèvremont et al. (2013) (529 ± 3 Ma) in d.

The Adrar-n-Takoucht formation was mainly sampled in both flanks of the anticline of Adrar-n-Takoucht (13 sites), located in the southern part of the Sirwa inlier (Figs. 1b and 2). This anticline was formed during the emplacement of the Adrar-n-Takoucht formation during the Ediacaran (Blein et al., 2014b). One volcanic layer was sampled on both flank of this fold corresponding to the sites 11 and 12. This formation was also sampled in the Agadir Melloul inlier in one site (78).

The Tadoughast and Fajjoud formations were sampled in tuffs and highly welded tuffs of trachytic composition, in the Agadir Melloul inlier (64 sites), in the Iguerda inlier (3 sites) and in the Sirwa inlier (1 site) (Figs. 1c and 2). Two sites were also sampled in the Bou Azzer and Saghro inliers on trachytic tuffs correlated respectively to the Fajjoud and Tadoughast formations (Figs. 1d and 2). The Tadoughast and Fajjoud formations were mainly sampled in an anticline and syncline located in the Agadir Melloul inlier (48 sites). In this folded structure, seven volcanic layers, comprising both tuffs and highly welded tuffs, were sampled on several sites from both flanks. This fold was formed during the Hercynian orogeny (Souleimani et al., 2013). The clasts of an intraformational conglomerate interstratified with the summital massive pyroclastic flows of the Tadoughast Formation were also sampled (site 62).

The Djebel Boho formation was sampled in the southern and northern flanks of the Bou Azzer inlier which was folded during

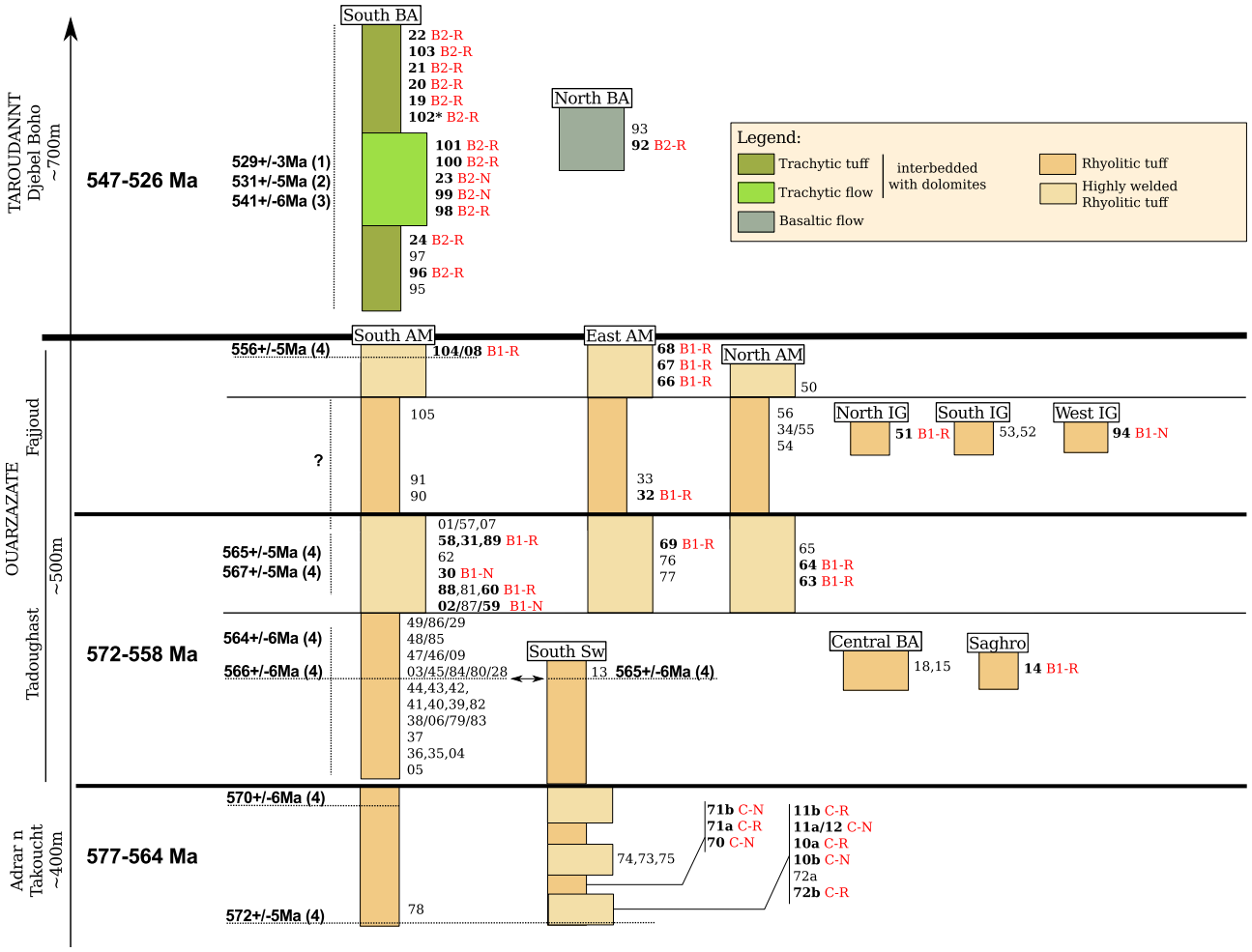


Fig. 2. Stratigraphic columns of the Ouarzazate and Taroudannt groups and stratigraphic position of the sampling sites. Site numbers from the same volcanic layer are separated by a "/". Those in bold carry either the components B or C, in reverse (R) or normal (N) polarity. Abbreviations mean: BA: Bou Azzer, AM: Agadir Melloul, IG: Iguerda, SW: Sirwa. U-Pb age comes from: (1) Ducrot and Lancelot, 1977, recalculated by Chèvremont et al. (2013); (2) Gasquet et al., 2005; (3) Blein et al., 2014a; (4) Blein et al., 2014b. Indicative thickness for each formation is given in meters.

the Hercynian orogeny. In the south, tuffs and volcanic flows of trachytic composition were sampled in a monocline (15 sites). In the north, 2 sites were obtained from basaltic flows.

3. Methods

Paleomagnetic cores were oriented using both magnetic and sun compasses whenever possible. For a total of 793 paleomagnetic cores sampled, 416 cores were oriented with both techniques. The average difference between the azimuths calculated with the sun and magnetic measurements is low (mean = 0.22°, σ = 1.86), supporting the absence of large errors in the orientation of the cores. In laboratory, these cores were cut into cylindric specimens of 22 mm high and 25 mm wide. Paleomagnetic measurements and demagnetizations were conducted in the Paleomagnetic laboratory of Institut de Physique du Globe de Paris (IPGP). Magnetizations were measured using a JR-6 from AGICO, thermal demagnetizations were performed with a magnetically shielded furnace with an internal field less than 20 nT up to 700°C and alternative field (AF) demagnetizations were conducted using a LDA-3F from AGICO up to 100 mT. All the demagnetization procedure was carried out in a shielded room with internal field less than 1000 nT.

Demagnetizations conducted on pilot samples showed that thermal demagnetization was the best procedure to isolate the paleomagnetic components in our samples. Therefore, most of samples were treated using this technique. During thermal treatment, bulk susceptibility was controlled to identify possible chemical or mineralogical changes due to experimental heating. In order to assess the degree of internal deformation which can occur in pyroclastic deposits during their emplacement (Ellwood, 1982; Incoronato et al., 1983) or due to later tectonic deformations, we routinely measure the anisotropy of magnetic susceptibility (AMS) in our samples with a KLY-3 from AGICO before starting demagnetization.

Processing of paleomagnetic and AMS data was carried out with the Paleomac software (Cogné, 2003). Paleomagnetic results were visualized in orthogonal (Zijderveld, 1967) and stereographic projections; directions were computed using principal component analysis (Kirschvink, 1980) and site-mean directions were calculated with Fisher statistics (Fisher, 1953). In case of overlap between two magnetic components, remagnetization great circles were used (Halls, 1978).

The magnetic behavior of our samples was also studied in high field using a vibrating sample magnetometer (VSM) up to 1.8 T. We performed hysteresis loops on one sample per site. The magnetic carriers were identified using both the hysteresis loops and the evolution of the magnetic moment obtained during demagnetization. The structure and aspect of the magnetic carriers in the main lithologies were investigated by scanning electron microprobe

(SEM). In-situ analysis of the chemistry of our samples was carried out using an EDS-X probe.

4. Results

4.1. Magnetic behavior

The natural remanent magnetizations (NRM) for all lithologies are comprised between 10E−4 A/m and 10E−1 A/m and can reach 1 A/m in some samples of ignimbritic deposits and volcanic flows. During demagnetization, a first decrease of 10–50% in the magnetic moment is generally observed at low to intermediate temperatures (<440°C). Most of the paleomagnetic directions isolated on this demagnetization range are of positive inclinations and their in-situ mean is: D: 355.1°, I: 46.6°, n: 472 samples, k: 13.8, α_{95}: 1.8, oriented close to the direction of the current geomagnetic field (Fig. 3a). After bedding correction, the mean direction is: D: 355.7°, I: 45.0°, k: 2.9, α_{95}: 6.0° (Fig. 3b). The fold test of McFadden (1990) is negative at 95% and 99% of confidence level which suggests that this component is a post-tectonic remanence. The negative fold-test and the proximity of this component in in-situ coordinates to the current geomagnetic field strongly suggests that this component is a recent remagnetization.

At higher temperature, three magnetic behaviors were identified. In the first type, a sharp decrease of the magnetic moment is observed between 650 and 690°C (Fig. 4a Type 1). This type of samples is not sensitive to AF-demagnetization, the remanent magnetization changing less than 10% up to 100 mT (Fig. 4b Type 1). This is indicative of minerals of high coercivity. Saturation of the remanence for these samples is reached at field higher than 500–1000 mT (Fig. 4c Type 1). These different experiments therefore suggest that hematite is the main carrier of the remanence.

In the second type, magnetic moment drops around 580°C, 90% of the NRM is destroyed at 30 mT and saturation of the remanence deduced from hysteresis loops is reached at 200–300 mT (Fig. 4a–c Type 2). All these experiments suggest that magnetite is the main carrier of the remanence in these samples.

The third type is a mixing in variable proportion of the two previous behaviors and is also the most observed in our sample collection. Two sharp decreases of the magnetic moment are observed at 580°C and 680°C indicative of magnetite and hematite (Fig. 4a). During AF demagnetization, magnetite magnetization is destroyed at field less than 100 mT (Fig. 4b). In some samples which were then heated up to 700°C, no decrease of the magnetic moment was observed at 580°C and the magnetization was totally destroyed at 680°C, indicative of hematite. Hysteresis loops for these samples display generally a wasp-waisted shape (Fig. 4c) which suggests the presence of minerals of two distinctive coercivities compatible with magnetite and hematite contributions.

Hematite as the main carrier of the characteristic magnetization has already been observed in pyroclastic rocks (Paquereau-Lebti et al., 2008; Roperch et al., 2006). In these studies, hematite can be observed with magnetite but hematite can also be the exclusive high temperature carrier in some cases (e.g. Paquereau-Lebti et al., 2008). Hematite is also observed in precambrian pyroclastic rocks as for example in the study of the Ediacaran ignimbrite of Campo Alegre (D'Agrella-Filho and Pacca, 1988). Consequently, the mixture of hematite and magnetite found in this study is not in disagreement with a primary origin.

4.2. SEM observations

SEM observations were conducted on seven representative samples (mostly of type 3) from all the different lithologies collected in this study. In all the lithologies, we identified micrometric to submicrometric titanium-iron oxides with a low Ti content (<10%) which could be the magnetic carriers of high temperature observed in the rock magnetism experiments (Fig. 5a and b). In most of the samples, these minerals display homogeneous textures and angular shapes (Fig. 5b). Exsolutions of Ti/Fe are also observed in larger grains (>1 μm) which are typical of primary volcanic crystallizations (Fig. 5c). In samples of rhyolitic tuff from the Tadoughast Formation, most of the oxides are submicrometric and show acicular textures, agglomerated in clusters. The contour of these clusters delimits the fingerprint of a primary iron oxide which has been altered and replaced by the acicular iron oxides. Therefore, the acicular iron oxides are of secondary origin and probably dominate the remanence in these samples which indicates that this lithology has been more altered than other lithologies.

4.3. Paleomagnetic results

At high temperature (>440°C), three components were identified, independently of their magnetic behaviors (A, B and C). Anisotropy intensity (defined by Jelinek, 1981) is less than 5% for 96% of

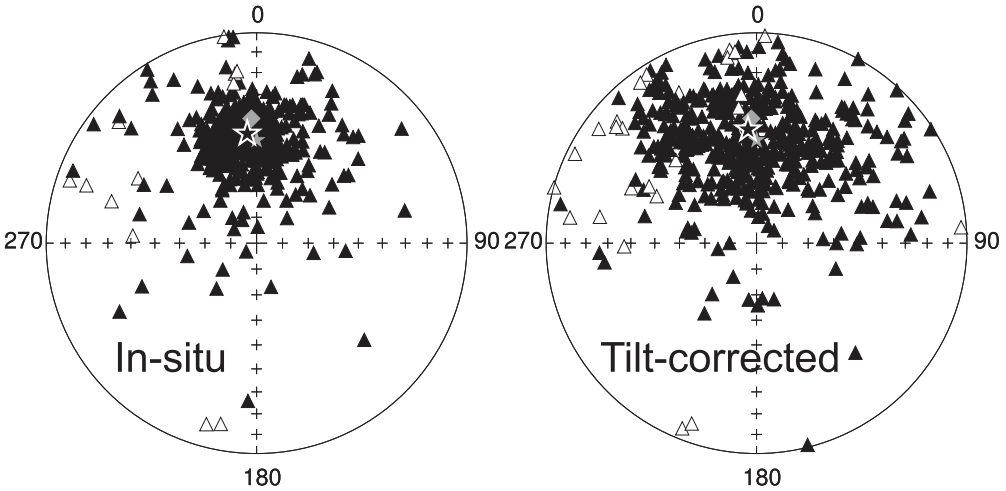


Fig. 3. Low to medium unblocking temperature directions identified in 472 samples from all studied geological formations and presented in in-situ (left) and tilt-corrected (right) coordinates. Directions are plotted in equal-area projection where open (full) triangles represent directions pointing up (down). The black star indicates the Fisher mean calculated with all these directions. The grey star and diamond correspond respectively to the Earth's magnetic field of January 2014 and the mean dipole calculated from the IGRF model (Finlay et al., 2010).

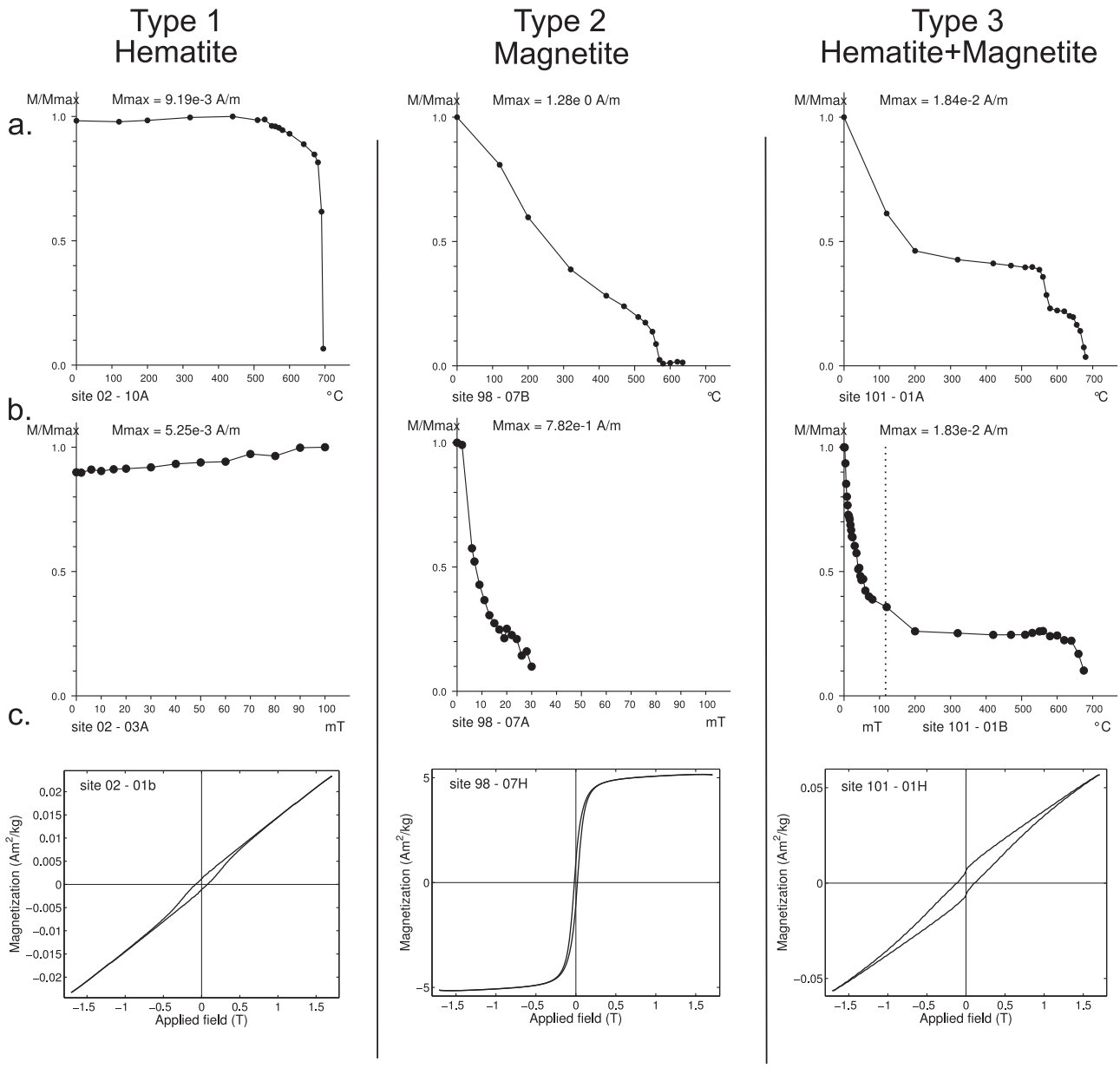


Fig. 4. Three typical magnetic behaviors observed in the studied samples. a. Thermal demagnetization curves. b. Alternating field demagnetization curves. c. Hysteresis loops.

the studied samples and do not exceed 7% for all the samples. These intensity values are typical of undeformed recent ignimbrites (Le Pennec et al., 1998; Paquereau-Lebti et al., 2008). Anisotropy intensity is similar in tuffs and highly welded tuffs, suggesting that there is no large influence of the degree of welding on the magnetic fabric. Therefore, we consider that the degree of deformation in our samples is relatively low and we did not applied corrections to the remanence.

4.3.1. Component A

The component A consists of low inclination (<30°), south-east declination single polarity directions (Fig. 6). These paleomagnetic directions are identified between 440 and 680°C (Fig. 6a–d) and can be slightly different from high temperature (between 440°C and 600°C) to very high temperature (>600°C) (Fig. 6c–d). This component is broadly observed in the Adrar-n-Takoucht, Tadoughast and Fajjoud formations and is only present in few samples of the Djebel Boho formation. This component is especially observed in the tuffs and much less in the highly welded tuffs.

This component has been identified in five volcanic deposits of the Tadoughast and Fajjoud formations, sampled in several sites in the anticline/syncline in the Agadir Melloul inlier. Therefore, we were able to perform layer-scale fold-tests for these directions. This kind of tests is particularly useful in the study of volcanic rocks as it allows us to ignore potential effect from the secular variation. The best grouping for each layer was obtained at different unfolding rate, from 100% to 40% (see two examples in Fig. 6e and f), suggesting that this component was acquired before and during the folding event related to the Hercynian collision. This strongly suggests that this component is of secondary origin and is linked to the folding process. At regional scale, the best grouping of the site-mean directions identified in this study, is obtained for an unfolding rate of 40% (D: 128.0°, I: 8.8°, N: 53 sites, k: 13.3, α_{95}: 5.6, Fig. 7a and b) and is consistent with the layer-scale fold-test results. The global means calculated with the magnetite-bearing,

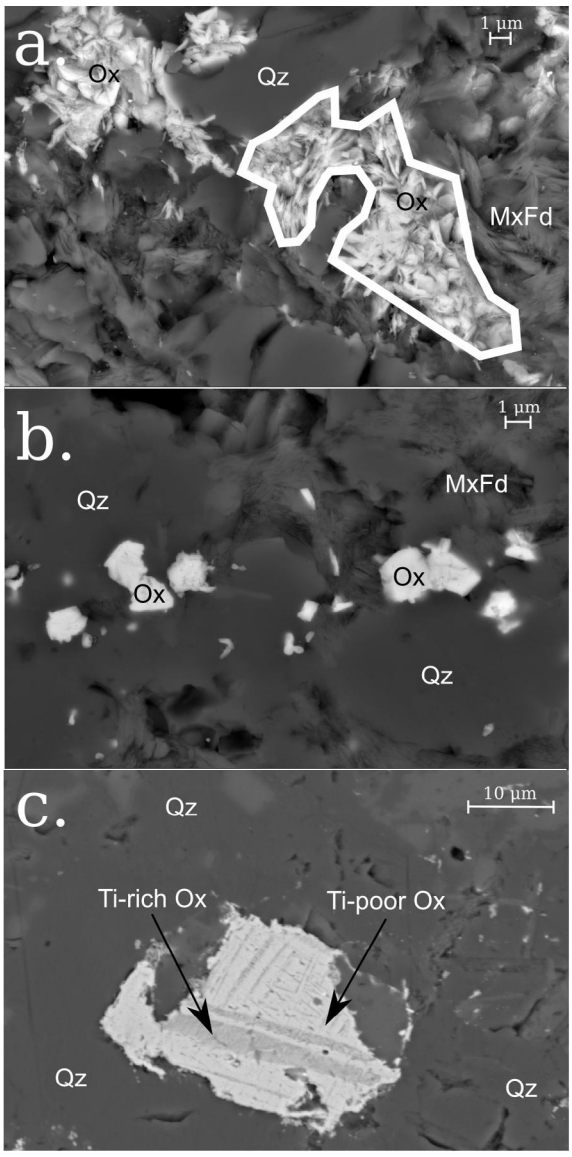


Fig. 5. Back-scattered electron SEM images obtained on representative samples of tuff (a.) and highly-welded tuff (b. and c.) of rhyolitic composition. In a., the white contour delimits the fingerprint of a primary iron oxide which was altered and replaced by acicular iron oxides. Abbreviations are: Ox: Iron oxides, Qz: Quartz, MxFd: matrix of feldspath composition.

hematite-bearing or mixed-bearing site-mean directions do not show any statistical differences (Table 2), suggesting that there is no influence of the magnetic carrier on the acquisition of the direction. For the calculation of the component A, we used the site-mean direction having the best α_{95} from each site (Table 1).

4.3.2. Component B

The component B is a dual polarity high inclination direction when corrected for bedding (Fig. 8a–f) found in the Taddoughast, Fajjoud, and Djebel Boho formations. This direction is most often identified at temperature >600°C indicating a mineral of the titanohematite family (Fig. 8a–e). In some cases, a component of same direction is determined in a range comprised between 500 and 600°C, which may indicate the contribution of titanomagnetite as a magnetic carrier for this component (Fig. 8c and f). The component A previously discussed is sometimes also observed, and can be easily separated, thanks to its different inclination. An example is given with the Zijderveld plot shown in Fig. 8b and d (represented after tilt correction): the component A is determined between 320 and 550–600°C while the component B is determined at higher temperature. The best characterized component B is observed almost exclusively in the highly welded tuffs of the Tadoughast and Fajjoud formations on one hand, and in the tuff and lava flows of the Djebel Boho formation.

The component B has been identified in two highly welded tuffs from the anticline/syncline of Agadir Melloul in the Tadoughast and Fajjoud Formations which were sampled in several sites, allowing us to perform a fold-test at the layer level. In the layer corresponding to the sites 08 and 104, the component B has been identified at very high temperature (>650°C) in both sites and in the same polarity. Best grouping of these high positive inclination site-mean directions is obtained at an unfolding rate of 100% which suggests that this component was acquired before the folding event (Fig. 8e and f). A similar result is obtained for the layer corresponding to the sites 02 and 59 in opposite polarity (Fig. 8g and h). These high negative inclination site-mean directions group at an unfolding rate of 100%. The component B was found in the late Ediacarian Fajjoud and Tadoughast sections (dated at around 560 Ma), as well as in the early Cambrian Djebel Boho formation, (dated at around 535 Ma). The two lower formations are separated from the upper one by some 25 Ma, and justify a separation into two directional groups based on component B, the group B1 for the lower part of the section and B2 for the upper one.

The mean directions B1 observed in the Agadir Melloul and Iguerda inliers are presented in Fig 9a and b. The best grouping of the selected site-mean directions is obtained for an unfolding rate of 100% (N: 20 sites, n: 99 samples, D: 93.7°, I: 67°, k: 10.7, α_{95}: 10.5°; Fig. 9a and b) and is consistent with the previous layer-scale fold-test results. The fold-test of McFadden (1990) is positive at a confidence level of 99% which indicates that these directions were acquired before folding. We also added the results obtained in the sites 94 and 14, respectively from the Bou Azzer and Saghro inliers of the Ouarzazate group, to calculate the mean direction B1. A reversal test (McFadden and McElhinny, 1990) yielded an inconclusive result. The result of this test is probably biased by the low number of negative high inclination directions obtained from 2 sites only, that of course have not averaged the secular variation.

The clasts from an intraformational conglomerate of the Tadoughast formation were also studied. This conglomerate is intestratified with the summital highly welded tuffs of this formation where the component B was especially observed in the section. In five seventh of the clasts, there is an overlap of magnetic components (Fig. 10). The great circles, calculated from each of their demagnetization paths, converge to a direction similar to component A. The presence of this common direction in some clasts suggest that this conglomerate was remagnetized by the component A. On the contrary, very high temperature components are randomly distributed at a confidence level of 99% (test of Watson, 1956), and support a positive intraformational conglomerate test for these directions. The absence of the component B in the clasts of this intraformational conglomerate supports its primary nature.

In the Djebel Boho Formation, the precision factor of the site-mean directions do not change significantly after bedding correction (k_{IS}: 15.8, k_{TC}: 14.4; Fig. 9c and d). The fold-test of McFadden (1990) is negative at 95% (χ^2). However, the distribution of these directions is highly asymmetric, the component B being identified in respectively 10 sites in the southern and only one site on northern edges, respectively, of the Bou Azzer inlier (Figs. 1d and 2). We tried to overcome this asymmetric distribution by averaging the site-mean directions obtained in the southern edge of the Bou Azzer inlier in order to balance the fold-test. The best grouping of this mean with the site-mean direction from the

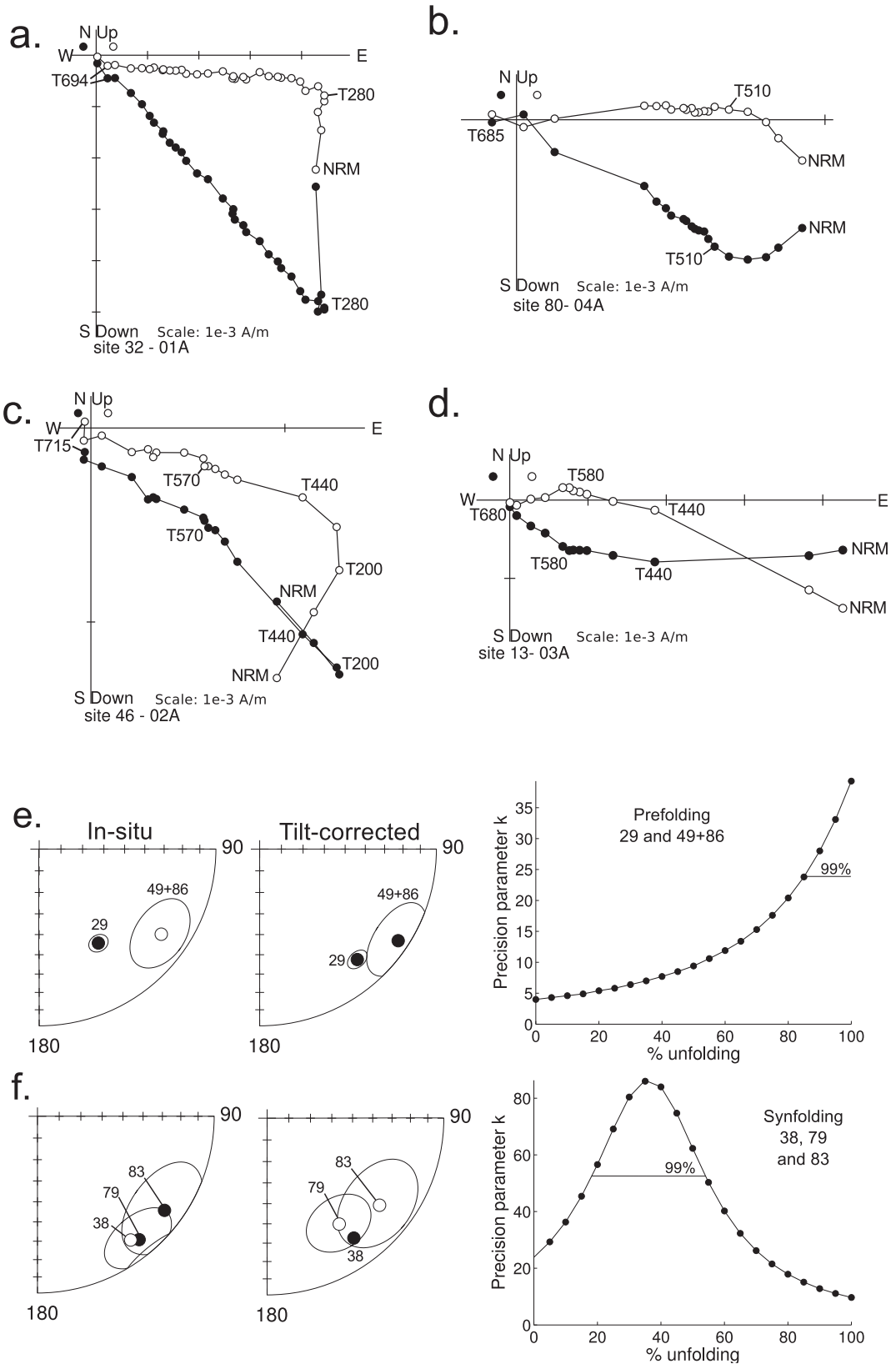


Fig. 6. Examples of demagnetization behaviors to thermal cleaning (a. to d.) of samples carrying the component A at high to very high temperature (>440°C) plotted in Zijderveld (1967) diagrams. W: west, N: North, E: East, S: South. Open (solid) symbols mean projection onto the vertical (horizontal) plane. Results are presented in tilt-corrected coordinates. Equal-area projections where open (full) circles represent directions pointing up (down) (e. and f.). In e. and f. are shown two examples of layer-scale fold-tests carried out in two tuffs of the Tadoughast Formation in the Agadir Melloul anticline/syncline. (left) In-situ coordinates, (middle) tilt-corrected coordinates, (right) evolution of the precision factor k with the unfolfing rate. The 99% confidence level for k about the best unfolding percentage is shown as a bar within the curve.

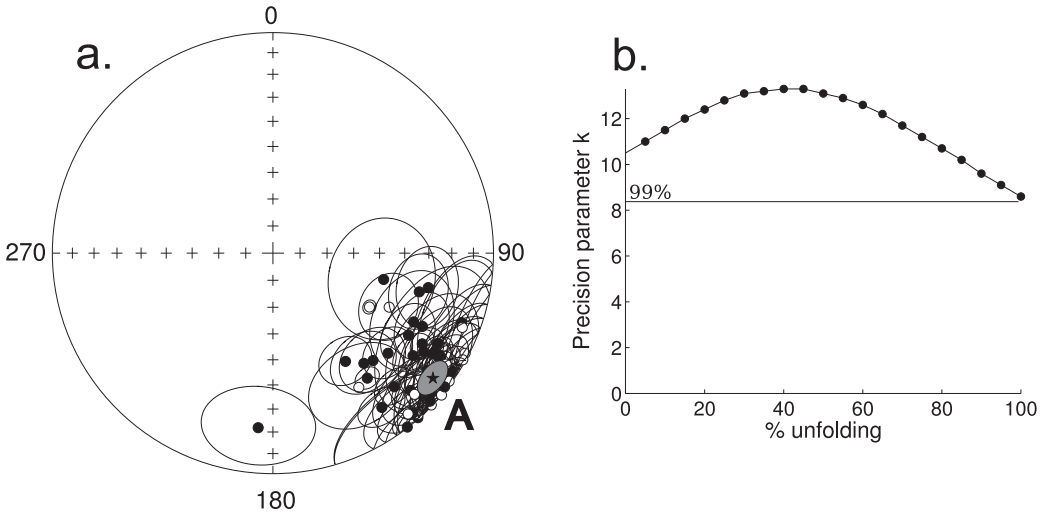


Fig. 7. a. Component A determined from site-mean directions identified in the Fajjoud, Tadoughast and Adrar-n-Takoucht formations at an unfolding rate of 40%. Directions are plotted in equal-area projection where open (full) circles represent directions pointing up (down). Only the site-mean direction having the best α_{95} from each site is represented. In b., evolution of the precision factor k with the unfolding rate for the site-mean directions presented in a. The 99% confidence level for k about the best unfolding percentage is shown as a bar within the curve.

northern edge is obtained at an unfolding rate of 100% (Fig. 9c and d right bottom) which suggest that this component is a pre-tectonic remanence. The reversal test (McFadden and McElhinny, 1990) conducted on all the site-mean directions yielded a negative result. The result of this test is probably biased by the low number of negative high inclination directions, obtained from 2 sites only and can be one more time an effect of a poorly averaged secular variation.

Perrin (1994) already observed a component similar to our component B in sedimentary and volcanic rocks from the Taroudannt group in the southern and northern edges of Bou Azzer inlier (component B, Perrin, 1994). These directions are in good agreement with those identified in this study and group in tilt-corrected coordinates (Fig 9c and d). The fold-test calculated with the site-mean directions identified in this study and those from Perrin (1994) is inconclusive. By including the directions from Perrin (1994) in the calculation of the flank-balanced fold-test presented above, we obtain a positive fold-test at 95% and 99% (McFadden, 1990), suggesting that this mean direction, that we call B2, was acquired before the folding event which occurred during the Hercynian orogeny.

To assess the dispersion consistency of the mean directions B1 and B2 with more recent results from the literature, we calculated the commonly used angular standard deviation or parameter S. We obtained a dispersion of 36.1° and 30.9° for estimated paleolatitudes of 60 ± 14.9° and 55.1 ± 15.6° for each mean direction respectively. These dispersions are much higher than those observed for the last 5 Myrs showing dispersion between 15° and 20° for comparable paleolatitudes (Merrill and McElhinny, 1983; Hulot et al., 2010). Nevertheless, the dispersion of the mean directions B1 and B2 is comparable to other volcanic distributions. For instance, Chenet et al. (2008) reported a paleomagnetic study on volcanic rocks from the Deccan trap. The spread in inclination and declination around the mean for 50 site-mean directions is around 40° (see Fig. 11a in Chenet et al., 2008), similar to what we observed for the directions B1 and B2. Moreover, there is no statistical difference for the mean direction B2 between the mean directions calculated with volcanic rocks and sedimentary rocks. This supports a well averaged secular variation and suggest a low influence of secondary contamination on this component. The large dispersion of the mean directions B1 and B2 could also suggest that the dispersion of the Ediacaran paleosecular variation was higher than for the last 5 Myrs, but this point needs a careful work on all the already published paleomagnetic results and is beyond the scope of this paper.

4.3.3. Component C

The component C, found only in the Adrar-N-Takoucht formation dated around 571 Ma, consists of dual polarity low to medium directions of north-east and south-west declination (Fig. 11). These paleomagnetic directions are identified at high temperature between 440 and 600°C (>600°C; Fig. 11a and b) and at very high temperature (>600°C; Fig. 11c and d). In 10% of samples, we have observed this component and the component A. In Fig. 11b, the component C is isolated between 470°C and 580°C and the component A is observed at temperature higher than 600°C. In the site 70, the component A was either not identified (Fig. 11d) or observed in the same demagnetization range than the component C (Fig. 11e). In this latter case, we used great circles to separate these two components. The component C is solely observed in the Adrar-n-Takoucht Formation, in both tuffs and highly welded tuffs from the southern part of the Sirwa inlier, and was never observed in the samples, cores or formations where component B was identified.

This component has been identified in one ignimbritic deposit from the anticline of Adrar-n-Takoucht sampled in two sites (11 and 12), allowing us to perform a layer-scale fold-test. In both sites, similar directions of medium inclination and north-east declination were identified in the same polarity (Fig. 12a and b). Best grouping is obtained at an unfolding rate of 80% which suggests that these directions were acquired at the early stage of the folding event. The regional-scale fold-test gave a positive fold-test at a confidence level of 95% (test of χ^2 of McFadden, 1990, Fig. 12c and d), consistent with the layer-scale fold-test and suggests that this component is a pre-tectonic remanence. The fold being formed during the Ediacaran, the component C has probably been acquired during or close to the emplacement of the volcanic series of the Adrar-n-Takoucht formation. The bedding attitudes in the sampled sites do not display a horizontal fold axis due to their proximity to the eastern tip of the Adrar-n-Takoucht anticline. The fold axis computed with the structural data obtained in the anticline is barely determined, yielding a plunge of 22.6° and a direction of 76.4° with an uncertainty at 95% of confidence level of 38°. The

Table 1
Site-mean directions identified at high to very high temperature in the samples from this study.

Site	Litho.	Carrier	Comp.	n/N	D	I	D*	I*	k	α_{95}	Strike	dip	Slat (°N)	Slon (°E)
Adrar-n-Takoucht formation														
10a	h.w. Tuff	hem	C	4/4	169.6	−14.5	193.1	−33.4	104.7	9.0	20.9	52	30.483	−7.798
10b	h.w. Tuff	mag	C	8/8	28.5	9.5	33.0	−0.1	26.9	10.9	20.9	52	30.483	−7.798
11a	h.w. Tuff	mix	C	6/6	338.8	43.6	357.7	29.9	13.6	18.8	320.0	28	30.482	−7.798
11b	h.w. Tuff	mix	C	4/4	244.0	−18.3	243.5	8.9	452.8	4.3	320.0	28	30.482	−7.798
12	h.w. Tuff	hem	C	4/5	5.4	9.3	1.1	41.5	85.8	10.0	108.2	33	30.476	−7.797
70	Tuff	hem	A	3/10	112.2	37.6	97.4	21.4	101.2	12.3	316.8	30	30.479	−7.791
70	Tuff	hem	C	5+5GC/10	40.5	70.8	44.0	40.8	46.8	7.4	316.8	30	30.479	−7.791
71a	Tuff	mix	A	4/9	144.1	34.7	127.4	30.3	60.3	11.9	325.3	26	30.479	−7.791
71a	Tuff	mag	C	4/9	216.0	−42.6	220.5	−17.7	9.0	34.1	325.3	26	30.479	−7.791
71b	Tuff	hem	A	2/3	124.4	7.4	123.3	−2.3	318.9	14.0	325.3	26	30.479	−7.791
71b	Tuff	hem	C	3/3	29.8	8.4	29.1	−15.1	55.7	16.7	325.3	26	30.479	−7.791
72a	h.w. Tuff	mix	A	10/10	119.6	13.7	120.9	−24.2	10.2	15.9	10.5	40	30.483	−7.798
72b	h.w. Tuff	mix	C	6/11	188.8	−15.2	199.2	−12.6	25.9	13.4	10.5	40	30.483	−7.798
73	h.w. Tuff	mix	A	4/4	122.4	0.8	116.3	35.8	26.4	18.2	236.5	39	30.473	−7.787
74	h.w. Tuff	mix	A	6/6	114.3	24.0	122.4	−9.0	257.1	4.2	89.5	63	30.471	−7.786
75	h.w. Tuff	hem	A	7/7	111.5	5.5	109.7	−29.6	9.7	20.4	33.5	36	30.465	−7.814
78	h.w. Tuff	hem	A	5/5	135.5	13.8	208.3	57.8	68.5	9.3	193.5	83	30.219	−7.816
Tadoughast formation														
01	h.w. Tuff	–	uns.	0/6	–	–	–	–	–	–	158.0	22	30.227	−7.826
02	h.w. Tuff	hem	B1	6/6	283.4	−61.2	290.2	−83.2	135.4	5.8	191.2	22	30.223	−7.825
04	Tuff	hem	A	7/8	123.8	−6.1	127.4	29.1	76.7	6.9	189.4	39	30.221	−7.822
04	Tuff	mag	A	4/8	136.8	−55.7	120.9	−21.1	80.9	10.3	189.4	39	30.221	−7.822
05	Tuff	hem	A	4/8	122.5	−53.2	109.6	−22.8	63.4	11.6	178.0	34	30.221	−7.822
06	Tuff	–	uns.	0/4	–	–	–	–	–	–	15.5	50	30.218	−7.813
07	h.w. Tuff	–	uns.	0/6	–	–	–	–	–	–	7.0	60	30.217	−7.811
09	Tuff	mag	A	6/6	124.0	20.2	134.5	−47.7	57.9	8.9	10.2	75	30.218	−7.812
13	Tuff	hem	A	5/6	121.1	33.2	122.2	−11.5	184.9	5.6	38.6	45	30.479	−7.772
13	Tuff	mag	A	6/6	97.3	44.8	106.9	4.0	25.3	13.6	38.6	45	30.479	−7.772
14	h.w. Tuff	hem	B1	5/6	306.1	53.8	318.6	63.2	72.2	9.1	3.1	12	30.825	−6.787
14	h.w. Tuff	mag	A	3/6	122.6	33.1	119.6	22.5	188.0	10.6	3.1	12	30.825	−6.787
15	h.w. Tuff	mix	n. s.	3/3	75.4	30.0	85.7	15.4	17.1	30.7	49.7	28	30.495	−6.918
18	h.w. Tuff	mag	n. s.	5/5	21.8	−17.8	18.7	−23.2	11.1	24.0	347.7	10	30.488	−6.919
28	Tuff	hem	A	5/7	127.9	44.0	133.1	12.3	69.6	9.2	56.9	33	30.236	−7.807
29	Tuff	mix	A	4/6	147.8	37.1	138.5	17.7	483.8	4.2	10.9	26	30.236	−7.806
29	Tuff	hem	A	6/6	168.9	−3.5	172.4	−12.6	257.2	4.2	10.9	26	30.236	−7.806
30	h.w. Tuff	hem	B1	6/6	191.5	−63.7	229.5	−54.0	244.3	4.3	8.2	24	30.235	−7.805
31	h.w. Tuff	hem	B1	6/8	25.0	78.9	63.8	68.2	67.6	8.2	2.6	15	30.236	−7.805
31	h.w. Tuff	mag	A	4/8	117.6	35.2	114.4	21.4	30.8	16.8	2.6	15	30.236	−7.805
36/35 comb.	Tuff	hem	A	4/4	126.3	−18.8	126.2	17.6	103.2	9.1	210.1	36.6	30.221	−7.822
37	Tuff	hem	A	4/4	110.1	−4.4	111.6	31.1	27.3	17.9	190.8	36	30.221	−7.822
39/38 comb.	Tuff	mix	A	4/4	139.5	−14.1	140.1	16.6	132.0	8.0	187.9	40.6	30.221	−7.822
41/40 comb.	Tuff	mix	A	3/4	143.2	−10.0	144.6	15.8	34.5	21.3	186.9	37.2	30.221	−7.823
42	Tuff	–	n. t.	0/2	–	–	–	–	–	–	187.4	43	30.221	−7.823
43	Tuff	hem	A	2/2	101.4	13.3	107.2	51.4	363.2	13.1	181.2	39	30.221	−7.823
44	Tuff	–	n. t.	0/2	–	–	–	–	–	–	186.2	36	30.221	−7.823
45	Tuff	mix	A	6/6	136.4	−20.1	133.6	4.9	34.2	11.6	185.7	32	30.221	−7.824
45	Tuff	hem	A	6/6	172.1	−6.2	170.9	1.9	33.4	11.8	185.7	32	30.221	−7.824
3/84 comb.	Tuff	hem	A	10/10	125	−59.0	114.9	−32.2	360.1	2.5	185.7	32	30.221	−7.824
47/46 comb.	Tuff	mag	A	4/4	132.3	−18.1	130.6	6.3	66.8	11.3	187.8	29.2	30.221	−7.824
48/85 comb.	Tuff	mag	A	5/6	118.2	−13.5	118.4	15.3	16.9	19.2	188.3	30.6	30.221	−7.824
49/86 comb.	Tuff	mix	A	8/12	124.9	−16.9	123.5	6.8	16.1	14.2	179.9	28.4	30.221	−7.825
57	h.w. Tuff	mag	A	3/6	145.7	−21.4	151.4	−38.9	55.6	16.7	29.4	20	30.226	−7.836
57	h.w. Tuff	hem	A	4/6	168.6	9.0	168.0	−4.3	29.0	17.3	29.4	20	30.226	−7.836
58	h.w. Tuff	hem	B1	5/5	143.0	74.4	130.5	55.1	121.9	7.0	29.9	20	30.227	−7.836
58	h.w. Tuff	mag	A	4/5	146.0	22.7	143.9	4.5	143.4	7.7	29.9	20	30.227	−7.836
59	h.w. Tuff	hem	B1	4/4	75.8	−72.4	320.7	−77.1	83.5	10.1	12.8	26	30.236	−7.805
60	h.w. Tuff	hem	B1	6/6	85.7	63.9	93.5	39.5	21.6	14.8	13.5	25	30.236	−7.805
62	Conglo.	hem	A	5/7	115.7	31.4	116.5	16.5	0.	11.0	32.5	15	30.235	−7.815
63	h.w. Tuff	mix	B1	5/6	121.6	43.7	112.7	52.8	39.7	12.3	248.2	12	30.313	−7.775
64	h.w. Tuff	mix	B1	4/6	93.4	35.5	84.7	39.5	22.8	19.7	249.4	12	30.313	−7.775
65	h.w. Tuff	–	uns.	0/4	–	–	–	–	–	–	266.3	9	30.314	−7.775
69	h.w. Tuff	hem	B1	3/8	139.5	61.6	201.7	74.0	81.3	13.8	194.5	26	30.258	−7.807
69	h.w. Tuff	hem	A	3/8	168.2	2.1	171.4	13.1	88.9	13.2	194.5	26	30.258	−7.807
69	h.w. Tuff	mix	A	6/8	134.2	3.9	137.9	26.2	110.2	6.4	194.5	26	30.258	−7.807
76	h.w. Tuff	–	uns.	0/4	–	–	–	–	–	–	193.5	54	30.255	−7.807
77	h.w. Tuff	mix	A	4/6	103.5	26.8	102.3	77.8	19.9	21.1	193.8	51	30.256	−7.807
79	Tuff	mix	A	4/4	140.5	10.5	145.7	−28.3	45.5	13.8	13.5	49	30.218	−7.813
80	Tuff	hem	A	4/4	124.8	40.2	120.2	−6.0	401.8	4.6	15.5	48	30.218	−7.812
81	h.w. Tuff	mix	A	4/5	196.1	23.1	172.9	11.0	28.5	17.5	13.5	67	30.217	−7.811
82	Tuff	hem	A	3/4	129.2	32.9	124.2	6.1	274.9	7.5	9.5	30	30.235	−7.807
83	Tuff	mag	A	4/4	126.5	11.5	127.7	−20.7	24.8	18.8	13.5	35	30.235	−7.808
87	h.w. Tuff	hem	n. s.	6/6	194.4	−77.0	126.2	−67.2	14.9	17.9	181.5	22	30.222	−7.825
87	h.w. Tuff	mag	A	6/6	130.2	−9.6	130.0	7.7	47.8	9.8	181.5	22	30.222	−7.825

(*continued on next page*)

Table 1 (*continued*)

Site	Litho.	Carrier	Comp.	n/N	D	I	D*	I*	k	α_{95}	Strike	dip	Slat (°N)	Slon (°E)
88	h.w. Tuff	hem	B1	5/7	45.2	61.2	358.4	69.2	50.9	10.8	182.5	22	30.222	−7.826
88	h.w. Tuff	mag	A	7/7	133.4	−4.2	134.4	12.3	255.6	3.8	182.5	22	30.222	−7.826
89	h.w. Tuff	mag	A	7/8	135.7	12.7	142.7	28.4	78.9	6.8	179.5	24	30.224	−7.826
89	h.w. Tuff	hem	B1	4/8	92.2	35.9	93.9	59.9	16.0	23.7	179.5	24	30.224	−7.826
Fajjoud formation														
8	h.w. Tuff	hem	B1	7/8	272.2	63.3	119.4	55.3	33.5	10.6	17.4	60	30.215	−7.809
8	h.w. Tuff	mag	A	5/8	135	11.1	144.5	−41	31.2	13.9	17.4	60	30.215	−7.809
8	h.w. Tuff	mag	A	4/8	157	61.1	129.3	9.6	27.7	17.8	17.4	60	30.215	−7.809
32	Tuff	mix	A	7/7	138.6	−2.4	138.5	5.2	283.3	3.6	246.5	8	30.257	−7.810
32	Tuff	hem	B1	2+2GC/7	76.3	34.5	70.8	35.5	42.2	16.6	246.5	8	30.257	−7.810
33	Tuff	mag	A	5/5	133.8	−3.8	133.8	4.2	297.7	4.4	228.5	8	30.258	−7.811
34/55 comb.	Tuff	mag	A	3/11	125.2	18.4	120.0	20.2	247.9	7.8	305.0	16	30.325	−7.762
34/55 comb.	Tuff	hem	PEF	11/11	334.4	33.6	340.0	25.8	26.9	9.0	305.0	16	30.325	−7.762
50	h.w. Tuff	hem	A	4/4	132.1	4.9	130.4	4.2	153	7.5	313.3	21	30.327	−7.761
51	Tuff	hem	B1	3+3GC/7	217.8	81.4	125.2	85.3	110.3	6.9	335.6	10	30.285	−7.616
51	Tuff	mix	A	7/7	143.3	36.4	136.5	33.7	27.5	11.7	335.6	10	30.285	−7.616
52	Tuff	hem	n. s.	6/6	249.2	−1.7	249.2	4.4	41.7	10.5	31.5	10	30.187	−7.528
53	Tuff	hem	A	4/4	138.5	4.0	139.4	4.1	72.1	11.1	139.6	12	30.215	−7.575
54	Tuff	hem	PEF	6/6	358.1	49.6	7.9	39.7	500.2	3.0	317.5	14	30.324	−7.761
54	Tuff	hem	A	6/6	120.3	17.6	116.6	12.9	9.0	23.6	317.5	14	30.324	−7.761
56	Tuff	hem	A	4/4	125.6	4.5	123.7	6.3	453.9	4.3	299.5	20	30.326	−7.761
66	h.w. Tuff	mix	B1	6/6	111.3	42.6	110.8	54.6	142.6	5.6	203.1	12	30.282	−7.772
66	h.w. Tuff	hem	A	3/6	121.3	9.0	121.8	20.9	73.6	14.5	203.1	12	30.282	−7.772
67	h.w. Tuff	hem	B1	5/6	110.1	38.4	98.7	52.7	32.0	13.7	232.4	18	30.282	−7.773
68	h.w. Tuff	hem	B1	4/4	129.4	58.6	112.8	75.4	75.2	10.7	233.8	18	30.282	−7.773
90	Tuff	hem	A	4/4	130.1	5.5	130.1	−8.5	124.2	8.3	40.5	14	30.229	−7.831
91	Tuff	mix	A	4/4	128.3	7.5	128.3	−7.5	106.4	8.9	56.3	15.7	30.229	−7.830
94	h.w. Tuff	hem	B1	4/4	205.1	−80.8	132.0	−85.4	139.9	7.8	144.8	9	30.545	−6.998
104	h.w. Tuff	hem	B1	4/4	109.2	51.5	104.4	33.2	38.4	15.0	0.9	19	30.225	352.170
104	h.w. Tuff	mix	A	4/4	123.1	19.3	121.1	3.0	244.8	5.9	0.9	19	30.225	352.170
105	Tuff	mix	A	4/4	128.1	18.8	126.0	3.2	107.0	8.9	2.1	19	30.226	352.169
Djebel Boho formation														
19	Tuff	–	uns.	0/4	–	–	–	–	–	–	128.6	10	30.387	−6.663
20	Tuff	hem	B2	4/4	72.8	31.0	71.7	36.8	59.0	12.1	177.8	6	30.387	−6.664
21	Tuff	hem	B2	6/8	68.3	53.7	76.0	56.3	34.8	11.5	97.5	6	30.386	−6.664
22	Tuff	mix	B2	7/7	17.6	57.3	15.1	67.2	58.8	7.9	113.8	10	30.386	−6.664
23	lava flow	hem	B2	4/4	306	−59.7	315.3	−51.9	129.7	8.1	79.8	10	30.401	−6.660
24	Tuff	mix	B2	6/8	89.6	56.3	112.9	57.6	27.1	13.1	105.8	15	30.409	−6.655
92	lava flow	mag	PEF	4/4	346.9	48.9	346.8	37.9	35.5	15.6	255.9	11	30.521	−6.823
92	lava flow	mag	n. s.	4/4	236.8	20.2	241.0	23.5	98.4	9.3	255.9	11	30.521	−6.823
92	lava flow	hem	B2	4/4	126.1	80.9	41.6	83.0	306.9	5.3	255.9	11	30.521	−6.823
93	lava flow	hem	n. s.	4/4	198.8	31.7	204.1	42.7	67.8	11.2	259.3	13	30.521	−6.823
95	Tuff	–	uns.	0/4	–	–	–	–	–	–	123.7	28	30.410	−6.655
96	Tuff	mag	B2	4/4	37.7	63.5	19.3	88.4	39.1	14.9	128.9	25	30.411	−6.654
97	Tuff	–	uns.	0/4	–	–	–	–	–	–	128.9	25	30.411	−6.654
98	lava flow	mag	B2	4/6	49.3	67.1	151.6	81.9	57.3	12.2	121.1	26	30.400	−6.700
99	lava flow	hem	B2	4/4	291.8	−58.8	327.4	−54.7	764.5	3.3	120.0	23	30.404	−6.657
100	lava flow	mix	B2	5/5	115	87.1	190.2	78.6	17.5	18.8	114.9	11	30.399	−6.661
101	lava flow	hem	B2	5/5	55.9	34.5	60.6	43.9	141.2	6.5	116.9	11	30.397	−6.659
102	volc. Breccia	mix	B2	5/6	70.5	65.2	85.3	76.1	213.9	5.2	142.4	12	30.390	353.342
103	Tuff	hem	B2	5/5	57.2	51.3	64.9	65.6	41.3	12.1	132.9	15	30.386	−6.664
N1DU Perrin (1994)	dolomite	–	B2	6	174	72	84	81	75	7.5	290.8	20.3	30.51	−6.31
S2C2 Perrin (1994)	volc. + sed.	–	B2	7	41	70	75	73	104	5.7	72.4	11.5	30.41	−6.80
S3C2 Perrin (1994)	volc. + sed.	–	B2	2	52	54	60	76	122	0	136.5	22.2	30.37	−6.54

Comp: component, Litho: Lithology, Carrier: magnetic carrier referring to titanomagnetite (mag) titanohematite (hem) or a mixture of the two (mix), n: samples, D: declination, I: inclination, α_{95}: radius of the 95% confidence cone, k: precision parameter from Fisherian statistics. PEF: Present Earth field, n.s.: not selected, uns.: unstable. The site-mean directions with α_{95}>20° were added in the calculation of the components but do not change the results of the fold-tests. For some sites which are geographically close to each other (less than few meters) and/or for which only few samples were collected, we combined similar directions in order to improve the site-mean α_{95} (noted comb. for combined in the name of the site-mean directions).

* *Directions after bedding correction.*

application of this plunge does not change significantly the calculated mean direction for the component C (Table 2) and the results of the fold-tests remain the same. In this study, we decided not to take the plunge correction into account as the calculated plunge is very uncertain and does not seem to have a large influence on the results. A reversal test (McFadden and McElhinny, 1990) yielded an unconclusive result for the component C. The result of this test is probably biased by the relatively low amount of site-mean directions (less than 5 for the south-west polarity directions). When applied to the sample directions, this test is positive (class C) in in-situ and tilt-corrected coordinates. The angular standard deviation calculated for the component C is 23.9°. This value is slightly higher than for those observed for the last 5 Myrs but consistent with similar volcanic distribution of similar paleolatitudes (see for instance Chenet et al., 2008).

5. Discussion

5.1. Age of the magnetization

All poles calculated in this study are presented in Table 3. The component A is by far the most observed as it has been

Table 2
Mean remanence directions for the different magnetic components determined in this study.

Mean directions	Place	Geological Formation	N	n	D	I	k	α_{95}	D*	I*	k*	α_{95}*	S
A	AM + IG + SAG + SS	Fajjoud Tadoughast Adrar-n-Takoucht	53	254	128.8	9.2	10.5	6.4	128.0**	8.8**	13.3**	5.6**	16.8
A (mag only)	–	–	17	80	130.2	9.0	7.5	14.0	129.1	−0.2	9.7	12.1	–
A (hem only)	–	–	25	114	130.6	7.7	7.1	11.7	130.4	7.5	6.8	12.0	–
A (mix only)	–	–	18	83	134.1	11.3	10.1	11.4	133.3	12.0	9.1	12.1	–
B1	AM + IG + BA + SAG	Fajjoud Tadoughast	20	99	96.8	69.4	7.5	12.8	93.7	67.0	10.7	10.5	36.1
B2	BA	Djebel Boho	16	78	69.6	64.4	14.1	10.2	87.7	71.9	14.9	9.9	30.9
C	SS	Adrar-n-Takoucht	9	53	19.1	27.5	6.8	21.4	27.7	18.0	8.0	19.4	23.9
C plunge***	–	–	–	–	–	–	–	–	26.6	18.2	7.4	20.4	25.7

SS: South Sirwa inlier, AM: Agadir Melloul inlier, IG, Iguerda inlier, BA: Bou Azzer inlier, SAG: Saghro inlier. N: sites, n: samples, D: declination, I: inclination, α_{95}: radius of the 95% confidence cone, k: precision parameter from Fisherian statistics, S: angular standard deviation (calculated with virtual geomagnetic poles).
* Directions after bedding correction.
** Directions after bedding correction of 40%.
*** Plunge corrected using the following plunge axis: direction = 76.4°, dip = 22.6°, α_{95} = 38°.

significantly identified in all the four geological formations of the Ouarzazate group and also in few samples of the Djebel Boho formation. This component is a pre- to post-tectonic remanence and was acquired in close relation with the Hercynian main folding event. This secondary remanence was especially observed in the less densely welded pyroclastic deposits of rhyolitic composition. The comparison of these results with the apparent polar wander path of the Gondwana (Fig. 13; Torsvik et al., 2012) shows a good match of the pole A with the 300–330 Ma segment. This period of time corresponds to the folding events related to the Hercynian collision in the Anti-Atlas (e.g. Michard et al., 2008). This observation is thus consistent with the synfolding fold-test we calculated for this component. The component A was only identified in reversed polarity, which is consistent with the nearly 70 Myr reversed polarity state of the Earth magnetic field corresponding to the Kiaman superchron (Opdyke and Channell, 1996) during the largest part of the Permo-Carboniferous. Petrologic observations in the samples carrying the component A display recrystallizations of iron oxides. These secondary oxides acquired a chemical remanent magnetization probably in the direction of the component A. The chemical remagnetization mechanisms are generally associated to alteration by fluids which were either driven by tectonic processes or mobilized during heat burial (e.g. Elmore et al., 2012). The close match of the age of the component A and the Hercynian orogenic phase and the syntectonic acquisition of this component suggest that this remagnetization is linked to the Hercynian orogenic process. Migration of orogenic fluids have already been proposed in the literature as a mechanism for recrystallization. These fluids probably migrate along aquifers due to compression or by gravitational flow of meteoritic fluids from mountains (e.g. Oliver, 1986). This remagnetization was already observed in the North-West Africa and is consequently a large-scale event (Khattach, 1989; Perrin et al., 1988; Perrin, 1994; Boudzoumou et al., 2012). Rb-Sr, K-Ar and zircon fission track analyses conducted in the rocks of the Ediacaran and Cambrian formations in the Anti-Atlas yielded mean ages of 330–260 Ma (Charlot, 1976; Bonhomme and Hassenforder, 1985; Sebti et al., 2009), suggesting that temperatures as high as 350°C occurred during this tectonic event.

The component B was mainly observed in the highly welded tuffs of the Tadoughast and Fajjoud formations (corresponding to the pole B1) and in the tuffs and lava flows of the Djebel Boho formation (corresponding to the pole B2). This component is (i) observed in normal and reverse polarity in a large collection of samples (177 samples) from 36 sites distributed over 100 km (Agadir Melloul, Iguerda, Bou Azzer and Saghro inliers), (ii) is supported by positive fold-tests performed at the layer-scale in both polarities and at the scale of the geological formations, (iii) is not identified in the clasts of an intraformational conglomerate of the Tadoughast formation. Our B2 direction is moreover undistinguishable from the component B of Perrin (1994) for the same age, so that the early Cambrian direction is supported by both sediments and volcanic rocks. Therefore, we consider that this remanence is the characteristic component acquired during or just after the emplacement of the volcanic rocks. The age of the pole B1 corresponds to the age of the Tadoughast and Fajjoud formations, giving an age comprised between 572 and 551 Ma. The age of the pole B2 corresponds to the age of the Djebel Boho formation, giving a mean age comprised between 547 and 526 Ma. The Tadoughast and Fajjoud formations lie in some areas, on the paleoproterozoic basement (Blein et al., 2014b) and there are no indication of tectonic rotation with the overlying Djebel Boho formation. Consequently, the poles B1 and B2 are likely representative of the West African craton.

The component C was observed in all the lithologies studied in the oldest formation of the Ouarzazate group, the Adrar-n-Takoucht formation. This component is (i) a stable component of normal and reverse polarity observed in 53 samples from 9 sites and (ii) is supported by positive fold-tests performed at the layer-scale and at the scale of the geological formation which constrain its age before the Ediacaran fold. Therefore, we consider that this remanence is the characteristic component acquired during or just after the emplacement of the volcanic rocks of the Adrar-n-Takoucht formation, with a mean age of 571 ± 6 Ma. The Adrar-n-Takoucht formation is also lying directly on the paleoproterozoic basement (Blein et al., 2014b). Consequently, the pole C is likely representative of the West African craton.

One may recall the problem of hemispheric ambiguity: a positive inclination results from magnetization in the northern hemisphere during a normal-polarity interval or from magnetization in the southern hemisphere during a reversed-polarity interval, while the orientation of blocks is also critically determined by the choice of polarity. So it could be ambiguous whether our sites with positive inclinations are normal-polarity flows magnetized in the northern hemisphere or reversed-polarity flows magnetized in the southern hemisphere. Taking a reversed polarity for positive inclinations, the pole B2 from the Djebel Boho formation is situated close to a series of well dated poles of the Gondwana (Rapalini et al., 2015), and close to the 530 Ma pole of the Gondwana APWP of Torsvik et al. (2012). We chose this nearest distance approach for finally fixing WAC at high latitudes of the southern hemisphere. We have conserved the same polarity for the pole B1. The position of pole C is far from the poles B1 and B2 whatever polarity is chosen. The nearest position of the pole C with respect to poles B1 and

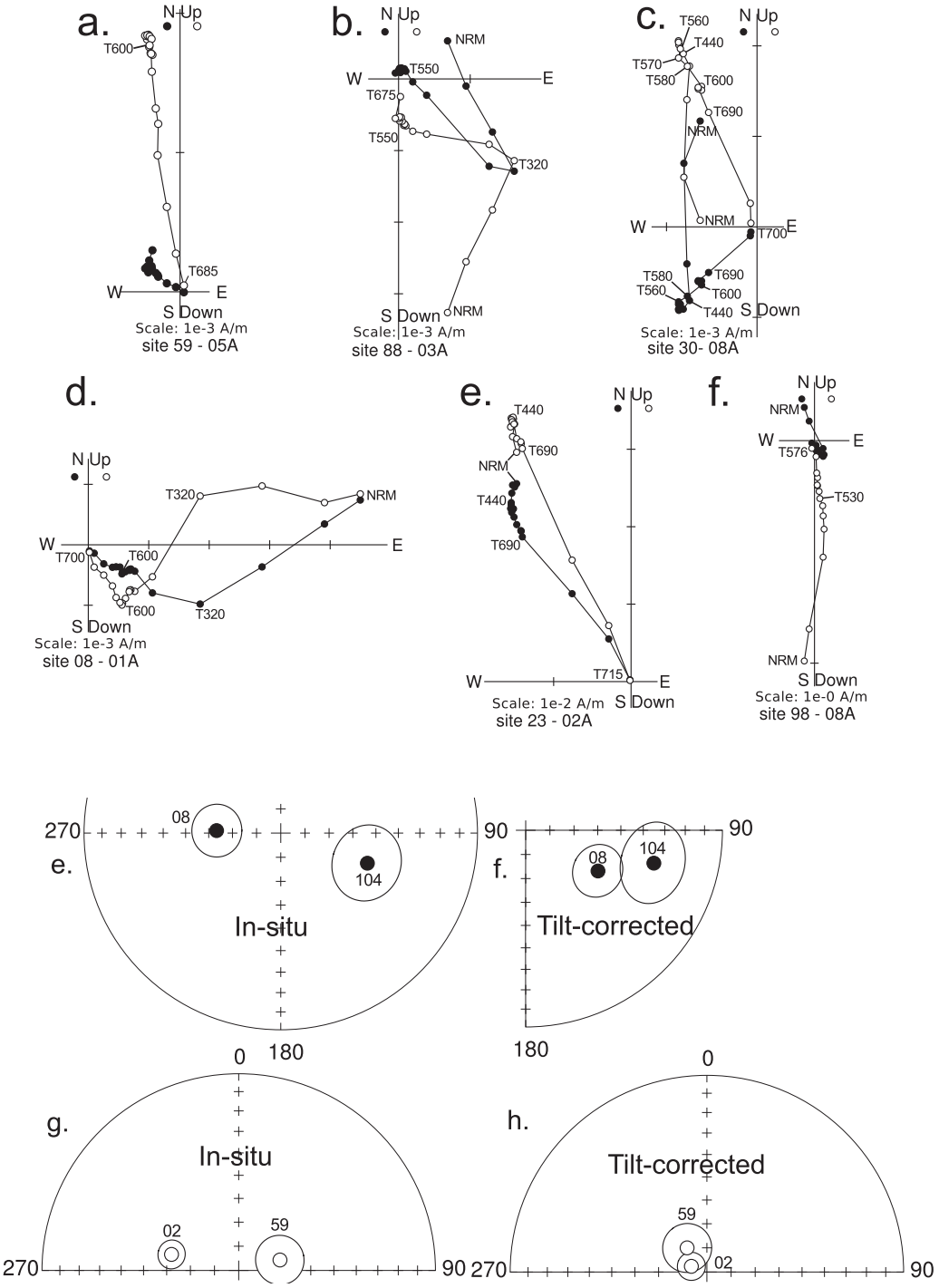


Fig. 8. Examples of demagnetization behaviors to thermal cleaning (a. and f.) of samples carrying the component B at very high temperature plotted in Zijderveld (1967) diagrams. W: west, N: North, E: East, S: South. Open (solid) symbols mean projection onto the vertical (horizontal) plane. These results are shown in tilt-corrected coordinates. In e. and f., and in g. and h., are shown the site-mean directions obtained in two highly welded tuff from the anticline/syncline of Agadir Melloul which were sampled in several sites. Equal-area projections show the result in in-situ (e. and g.) and tilt-corrected (f. and h.) coordinates.

B2 is obtained when the positive inclination directions are in reversed polarity, corresponding to an angular distance between C and B1 of 68°. Nevertheless, the pole C is in better agreement with other coeval Gondwana poles (see section 5.2.3) when the negative inclination is in reversed polarity. We thus chose this last solution.

The occurrence of normal and reverse polarities for the mean directions B1, B2 and C are indicated in Fig. 2 as a function of the stratigraphy. 7 reversals were observed in the Adrar-n-Takoucht formation. If the stratigraphic correlation between sites 94 and 14 and the south Agadir Melloul section is correct, 7 reversals are identified in the Fajjoud and Tadoughast formations. Finally, 2 reversals are observed in the Djebel Boho formation. In the Tadoughast, Fajjoud and Djebel Boho formations, 24 and 5 sites recorded respectively reverse and normal field whereas 4 sites of each polarity was found for the Adrar-n-Takoucht formation. For the Tadoughast and Fajjoud formations, the component B1 is more specially observed in the summital highly welded tuffs. In these horizons, we only observed a conglomerate (site 62) between the sites 31, 58 and 89 in reverse polarity and the site 30 in normal

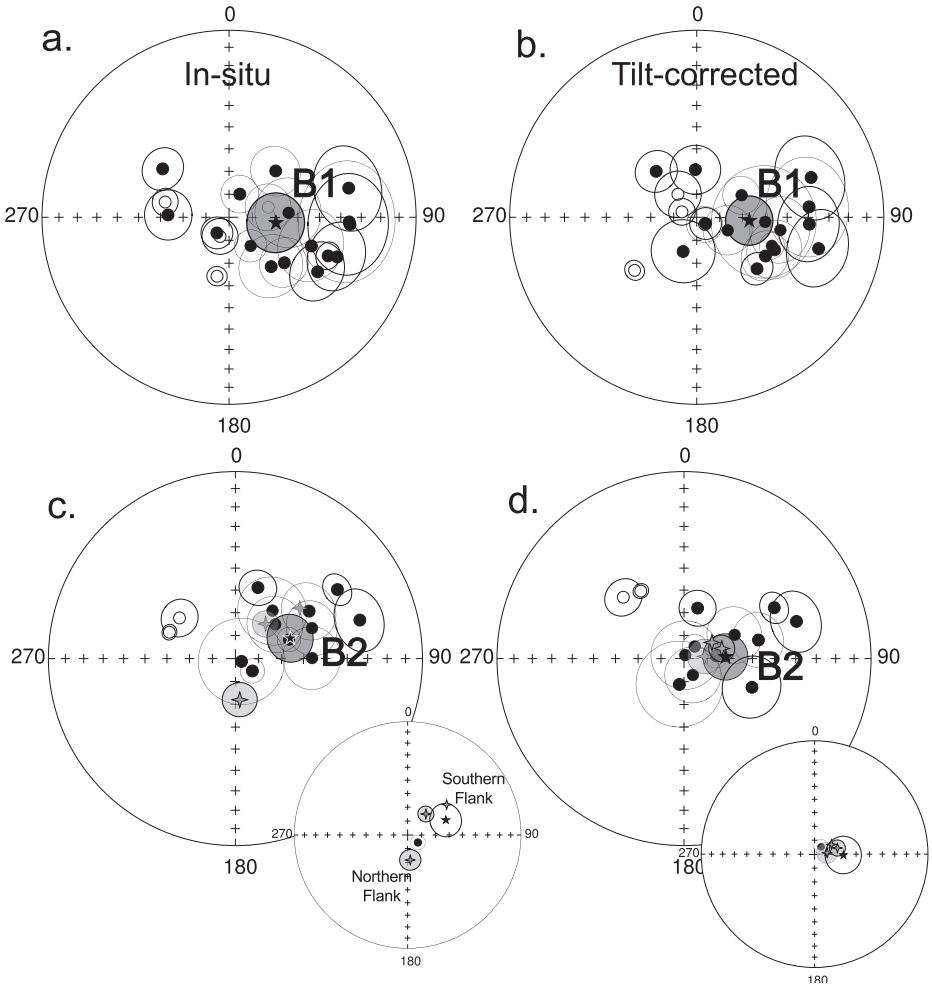


Fig. 9. Site-mean directions carrying the component B in the Tadoughast and Fajjoud formations, in in-situ coordinates (a.) and tilt-corrected coordinates (b.). The black star corresponds to the mean direction B1. Site-mean directions carrying the component B in the Djebel Boho formation, in in-situ coordinates (c.) and tilt-corrected coordinates (d.). The black star corresponds to the mean direction B2. On the bottom right of the figure c. and d., the directions from the southern and northern flanks of the Bou Azzer inlier are distinguished. The black star corresponds to the average direction of all the site-mean directions from the southern flank. The black point corresponds to the only site-mean direction from the southern flank (92). The 4-pointed grey stars correspond to the site-mean directions from Perrin, 1994. Directions are plotted in equal-area projection where open (full) circles represent directions pointing up (down).

polarity. Apart from this indicator, we do not observe sedimentation between each volcanic layer, which could indicate a short emplacement time. If so, it may explain why there is a dominance of reversed polarity. The south Bou Azzer section can be divided into three subsections. Two subsections of around 40 m thick comprise thin tuff horizons (pluridecimetric) interbedded with pluridecimetric dolomites levels, corresponding to the sites 22, 103, 21, 20 and 19 and the sites 24, 97, 96 and 95. The third subsection of around 250 to 300 m thick consists of metric to plurimetric lava flows interbedded with plurimetric dolomite horizons, corresponding to the sites 101, 100, 23, 99 and 98. Reversals are observed solely in the thicker subsection. If one assume that there is no important hiatus in these subsections, a longer time would have been needed to emplace the third section compared to the two other. Consequently, the likelihood to record a reversal in this section is also higher than in the two other small subsections.

5.2. Discussion of the paleomagnetic data from the West African craton

5.2.1. Selection criteria

In order to select the most reliable paleomagnetic results from the literature, we have used our own selection criteria. These criteria, which were modified from the classification of Van der Voo (1990), are presented as follows: (1) The age of the magnetization is not constrained by the age of other paleomagnetic poles and has to presumably be the one of the sampled unit. The unit age error has to be less or equal to 20 Ma. (2) The A_{95} has to be less than 16°. (3) Poles have to be calculated at least from 25 samples, distributed in at least 5 sites in case of volcanic rocks for having a good chance to integrate the secular variation. (4) Samples must be progressively demagnetized by thermal and/or AF methods (equivalent to demagcode≥4) and the direction of remanence has to be determined with a proper technique, using vector subtraction or principal component analysis. (5) There is no negative paleomagnetic test. (6) There is a structural continuity between the sampled region and the craton it represents. (7) There is no resemblance with younger poles, except if a paleomagnetic test demonstrates that a remagnetization of this age is not possible. We use these criteria in a more stringent way than does Van der Voo (1990). Only a pole passing all the criteria described above will be considered as a key pole to constrain the APWP, whereas the others will be eliminated or used with an extreme caution.

5.2.2. Apparent polar wander path of the West African craton (WAC)

The Gondwana is formed by many blocks amalgamated during several continental collisions occurring from the end of the Neoproterozoic to the early Cambrian (Fig. 14a, e.g. Cordani et al.,

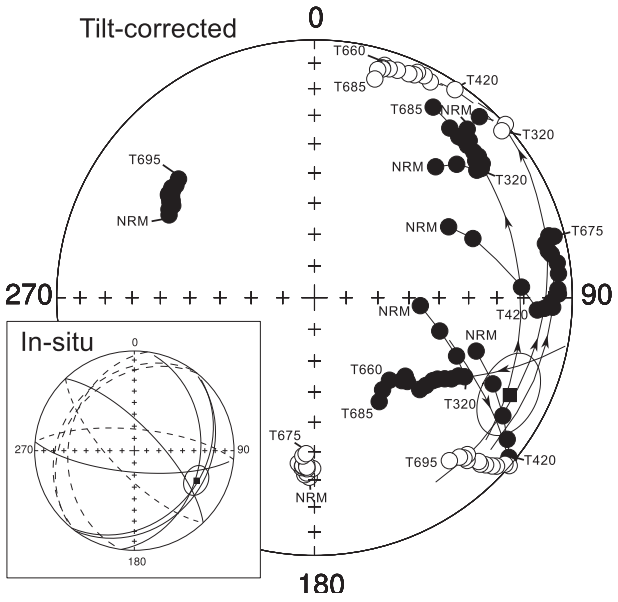


Fig. 10. Stereographic projection showing the thermal demagnetization path of 7 clasts from the conglomerate 62 in tilt-corrected coordinates. The arcs of great circles fitted to the demagnetization paths at high to very high temperatures are also represented. The averaged intersection is shown by a square with an ellipse of 95% confidence level. In the bottom-left, great circles and their intersection are represented in *in-situ* coordinates.

2013). The WAC is most generally associated with the Amazon craton, both forming a pre-Gondwanan continental entity. This connexion is suggested by geological similarities on both sides (Torquato and Cordani, 1981; Villeneuve and Cornée, 1994) and is supported by Paleoproterozoic to Mesoproterozoic similarities in their APWP's (Onstott and Hargraves, 1981; Nomade et al., 2003). In order to construct an APWP for the WAC, we started from the GPMDB database 4.6 (McElhinny and Lock, 1996) released in 2005 and added the most recent results from publications. The poles are shown in the Fig. 13 and detailed in Table 3. Poles WA1, WA2, WA4, WA8, A are close to the 270–330 Ma segment of the APWP of Gondwana (Torsvik et al., 2012) while poles WA3, WA5, WA6, WA7 and WA9 aligned between the late Paleozoic direction and the present Earth field (PEF). With the exception of pole WA1, no paleomagnetic test ascertains the primary nature of the magnetization. They are thus interpreted as remagnetizations acquired during the Hercynian orogeny, or a mixture of late Paleozoic directions and the PEF, as already proposed by Perrin (1994) and Tohver et al. (2006). The positive fold-test for WA1 was supposed to constrain the age of the magnetization before the Hercynian folding events (Khattach et al., 1995). Nevertheless, we have seen in our study that the late Paleozoic remagnetization could be acquired before or during the Hercynian folding event. Therefore, we suspect pole WA1 to be also a Hercynian remagnetization. WA5 and WA9 poles are based on blanket demagnetization and we consider that the magnetic components have not been accurately separated, which could explain their intermediary position. The poles WA5 and WA9 are also close to the poles B1 and B2 and may contain a similar direction. Nevertheless, we will not use these poles. WA12 has a very large 95% ellipse of confidence of some 40°, and is not selected either. The Adma diorite pole WA11 is determined by blanket method and is close to younger Cambrian poles. Nevertheless, a similar pole (WA10) is observed in the Tabankort rhyolites lying on the Adma diorite and give similar directions in the other polarity. Even if it does not prove the primary nature of the pole WA11, contaminations from secondary components are unlikely. WA11 and WA10 may define the APWP of WAC between 613 and 590 Ma but the age uncertainty of the Tabankort rhyolite is not given by Morel (1981) and a closer age to the one of WA11 is possible. WA10 is determined in only 3 sites and 8 samples from volcanic rocks. We thus chose to use only the most robust pole WA11 to define the early segment of the APWP of WAC.

Only few paleomagnetic data are available for the Amazon craton during the Neoproterozoic (Tohver et al., 2006). The only Ediacaran pole is determined from the Puga carbonates which are correlatives of the Marinoan glacial successions in Australia (Trindade et al., 2003), with an age of 635 Ma. This pole is close to the PEF in in-situ coordinates, and may refer to a very recent remagnetization. Font et al. (2005), based on rock magnetic experiments and petrologic observations, argue that the magnetic carriers of this direction are of primary nature which support its primary origin. Consequently, this pole may define the APWP's of WAC and Amazonia before 615 Ma (Fig. 13). The second component determined in the Puga carbonates may refer to a 540–520 Ma remagnetization acquired after the Paraguay Belt folding event (Trindade et al., 2003). We did not used it since its age is only constrained by the comparison with other paleomagnetic data. The final retained APWP is shown in Fig. 13, and displays two large polar motions, the first one between 615 Ma and 571 Ma and the second one between 571 and 562 Ma. This second event occurred between the emplacement of the Adrar-n-Takoucht formation and the Tadoughast formation dated at around 565 Ma. Therefore, this large polar motion ends as early as 565 Ma.

5.2.3. Comparison of the WAC APWP with other Gondwanan blocks

The West Gondwana is accreted to the central Gondwana blocks through the Transbrasiliano collision belt related to the closure of the Goias-Pharusian ocean (e.g. Cordani et al., 2013). The location of the suture is represented by a thick orange line in the Fig. 14a. The geological ages of collision spans from 620 to 580 Ma all along the belt (e.g. Caby, 2003; Cordani et al., 2013). The eastern part of the Gondwana is formed through two large-scale collisions. The East African orogen corresponds to the accretion of the Arabia-Nubia shield, India and Madagascar to the other Central Gondwanan blocks and occurs around 750–620 Ma (Meert, 2003). As this orogen is older than the period of interest in this study, the Central Gondwana will refer here to Sao Francisco-Congo, Hoggar, Arabia-Nubia shield, Madagascar and India (Fig. 14a). The second orogen, the Kuunga orogen, corresponds to the accretion of the Australian, Mawson and Kalahari cratons to the Gondwana and occurs from 570 to 530 Ma (Meert, 2003). The location of the suture is represented by a thick pink line in the Fig. 14a.

Over the last decade, a common APWP for the Gondwana blocks has been proposed from mid-Ediacaran to Cambrian times (Trindade et al., 2006; Tohver et al., 2006; Rapalini et al., 2015). This curve is constructed with poles from the Central Gondwana, Rio de la Plata and the Eastern Gondwana for the Cambrian segment. The APWP is represented by a black solid curve in the Fig. 14b. It goes from high-latitude in the southern hemisphere around 575–550 Ma to equatorial regions at early Cambrian times. The poles are presented in table 3. In the following discussion, we will only focus on the older segment of this APWP and will not include poles younger than 515 Ma from the compilation of Trindade et al. (2006) and Rapalini et al. (2015).

We first applied our selection criteria to their poles. Several poles pass all the section criteria and are considered as key data. They constitute two groups of poles of distinct age, at 575–550 Ma and at 525 Ma. The 575–550 Ma segment is constrained by 2 poles from the Central Gondwana corresponding to the Nola dyke pole (NO) and the Sinyai metadolerite pole (C3), and by one pole from the Rio de la Plata craton, the Sierra de las Animas volcanics pole (R2) (Fig. 14b.). The 525 Ma segment is constrained

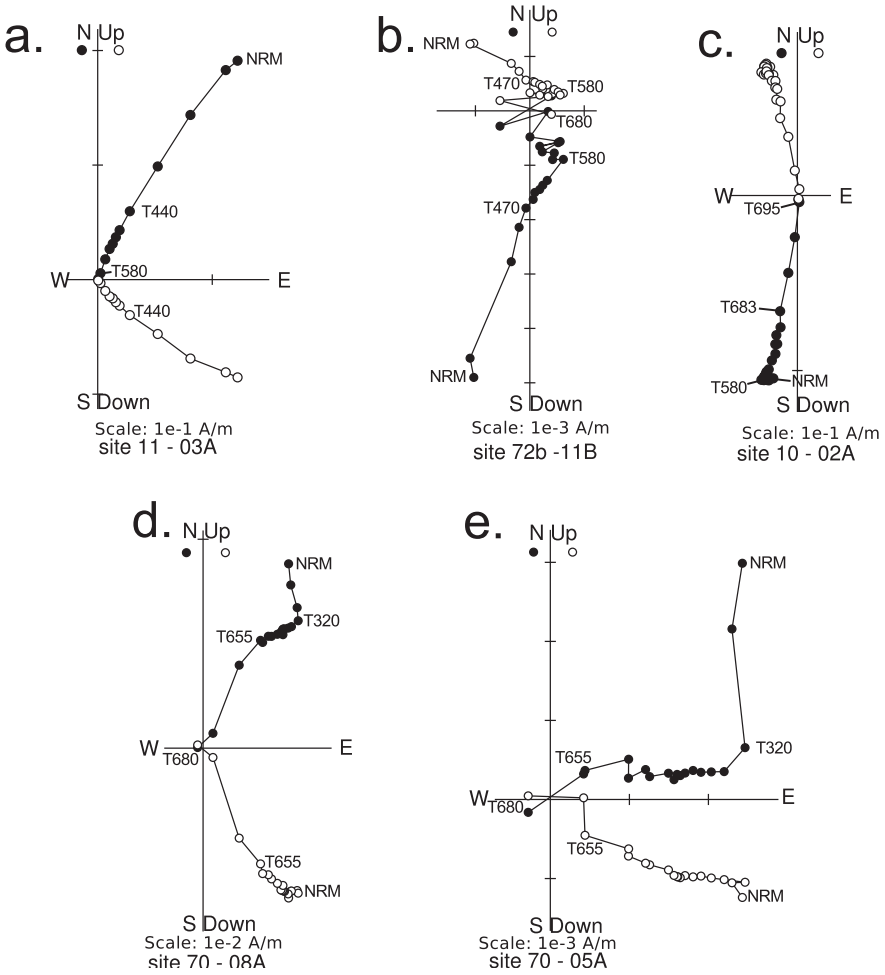


Fig. 11. Examples of demagnetization behaviors to thermal cleaning (a. to e.) of samples carrying the component C, plotted in Zijderveld (1967) diagram. W: west, N: North, E: East, S: South. Open (solid) symbols mean projection onto the vertical (horizontal) plane. These results are shown in tilt-corrected coordinates.

by 3 poles from the Central Gondwana corresponding to the Itabaiana dykes pole (C4), the Madagascar virgation zone pole (C6) and the Juiz de Fora Complex pole (C9), and 1 pole from the Eastern Gondwana, the Hawker group pole (A3). Most of the poles of lower quality are only dated by comparison with the APWP and do not satisfy our age criteria. This is the case for the poles SBc, SBd, P, C7 and C8 which are all close to the 525 Ma segment and may refer to poles of this age. Similarly, SBe and LB have been dated at 575 Ma as they are close to the 575 Ma segment. The pole R3 is ambiguous as it is observed in the Sierra de las Animas volcanics where R2, considered primary, was already identified (Sánchez-Bettucci and Rapalini, 2002). Trindade et al. (2006) and Rapalini et al. (2015) argued that this component is a Cambrian remagnetization as it is close to the 525 Ma segment of the Gondwana APWP. There is no paleomagnetic test to constrain its age and Rapalini et al. (2015) did not find this component in a new study of the Sierra de las Animas volcanics. Therefore, we considered that this pole is not reliable and was discarded from the discussion. The pole C2 from the Mirbat sediment group (Kempf et al., 2000) is defined from only 2 sites and 10 samples. Nevertheless, it is close to the good quality pole C3 of similar age and both may define a common segment around 550 Ma. However, the age of the Mirbat group was recently questioned, and an older age around 700 Ma (Rieu et al., 2006) was proposed. Consequently, we discarded this pole from our selection. For the Ntonya Ring pole (C5), most of the samples were not progressively demagnetized and these directions could be contaminated by other magnetic directions. Nevertheless, the close match of the corresponding pole with other coeval poles (C4, C6 and C9) support the absence of large contamination of these directions and we chose to retain this pole.

Between the 575–550 Ma and 525 segments, only few well-constrained poles are available. These poles come from the Eastern Gondwana only, corresponding to the upper Arumbera (A1) and Todd River (A2) poles. These two poles are suspected to have undergone regional rotations and may not be suitable to construct an APWP of the Gondwana. Indeed, large movements between the North and South Australian cratons may still occur during the early Cambrian (Myers et al., 1996; Wade et al., 2008; Li and Evans, 2011), continuing into the Paleozoic (Haines et al., 2001). Li and Evans (2011) estimated the Euler parameters of this relative movement based on a paleomagnetic analysis of Neoproterozoic poles of similar age from the Southern and Northern cratons. The application of this rotation to the Todd River (A1) and upper Arumbera (A2) poles moves them close to the Hawker group (A3) pole of similar age determined in the South Australian craton (Fig. 14d). A similar observation is obtained with the recent poles from the Arumbera sandstones (ARL and AECr; Mitchell et al., 2010) determined in the Northern Australian craton as well (Fig. 14d). Schmidt (2014) proposed an APWP for Australia based only on poles which belong to the South Australian cratons for the early Cambrian segment. This path, represented by a thick red line in the Fig. 14c, displays a large movement to the North between 609 and 590–560 Ma and then converges toward the Gondwana path during the lower Cambrian. However, the Ediacaran poles of

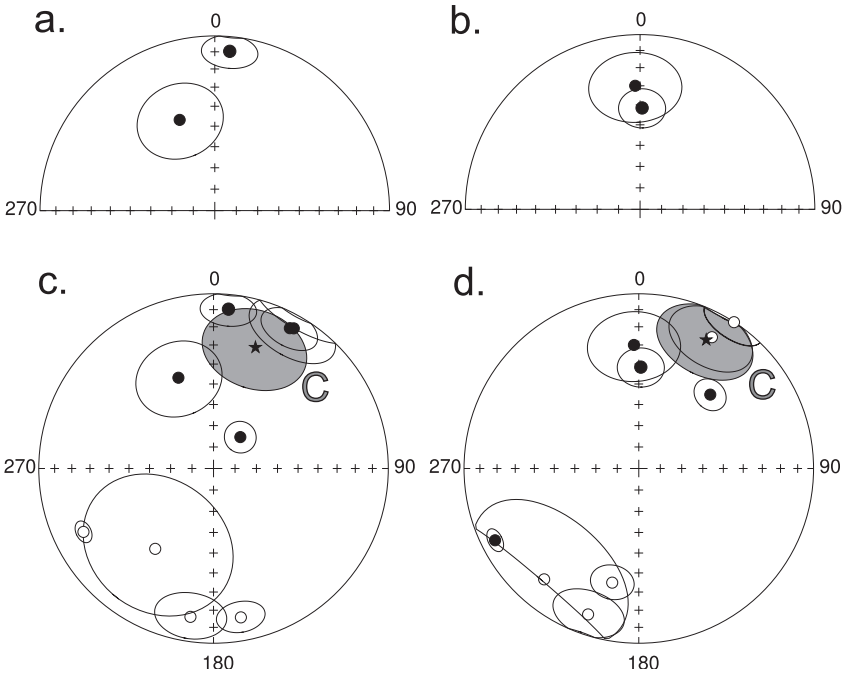


Fig. 12. Site-mean directions of component C determined in the sites 11 and 12 in in-situ (a.) and after bedding correction (b.). These two sites come from the same volcanic layer sampled on both flanks of the anticline of Adrar-n-Takoucht. Site-mean directions of component C determined in the Adrar-n-Takoucht formation are in in-situ (c.) and after bedding correction (d.). The Fisher mean is the star and is associated to a 95% confidence level ellipse colored in grey. Directions are plotted in equal-area projection where open (full) circles represent directions pointing up (down).

this APWP are ill-dated: a Rb-Sr isochron of 609 ± 64 Ma is obtained for the Brachina formation pole (BF) and a Rb-Sr isochron of 593 ± 32 Ma is obtained for the Bunyeroo formation pole (BU). The late Ediacaran to early Cambrian segment is better dated thanks to fossils of Ediacara and Cambrian fauna, leading to an age of 575–555 Ma for the Wonoka formation pole (WK) and of around 535 Ma for the Hawker group pole (HK). Once rotated, the North-Australian poles (A1, A2, ARL and AECr) are in good agreement with the 565–535 Ma segment of the APWP of Schmidt (2014), supporting the early Cambrian rotation between the Northern and Southern Australian cratons. If the poles from the North Australian craton are not included in the construction of the Gondwana APWP, there is no well-constrained poles between the 575–550 Ma and 525 Ma segments for East Gondwana.

Based on the high-quality data presented in Table 3, we defined the APWP's for the Central Gondwana, Rio de la Plata and East Gondwana (Fig. 14b). The APWP of East Gondwana converges toward the APWP's of Rio de la Plata and Central Gondwana at around 525 Ma, which is consistent with the youngest collisional ages of the Kuunga Orogen (Meert, 2003). Before 575 Ma, two well-constrained poles, dated around 595 Ma, are retained: the Campo Alegre pole (R1) from Rio de la Plata and the Dokhan volcanics pole (C1) from the Central Gondwana. Both poles are distinct, supporting two dissimilar APWP's for the Rio de le Plata and Central Gondwana before this age (Fig. 14b).

In the Fig. 14d, the APWP of the WAC proposed in this study is compared to the high-quality poles from the other Gondwana blocks. The poles B1 and B2, dated at 562 and 536 Ma respectively, are not far from the poles dated around 525 Ma from the Central and Eastern Gondwana and Rio de la Plata suggesting a small change in direction in the APWP for this period. The pole C, dated at 571 Ma, is in good agreement with the poles from the Central Gondwana and Rio de la Plata at around 575 Ma. If the West Gondwana is already accreted to the Central Gondwana and Rio de la Plata at 575 Ma as supported by geological data (e.g. Caby, 2003; Cordani et al., 2013), they should share a common APWP from 575 to 525 Ma. There is only one pole between 575 and 525 Ma for the Central Gondwana, referring to the Sinyai metadolerite (C3). This pole is close to the 575 Ma-poles and is in disagreement with the 565–535 Ma segment of the APWP of the WAC. Following the authors, this pole could be a remagnetization dated at 547 ± 4 Ma by the last opening of the 40Ar/39Ar system. The greenschist-facies metamorphic phase related to this event occurred between 570 and 530 Ma (Key et al., 1989) which means that the age of the remagnetization could be anytime between this period. The close match of this pole with the other 575 Ma-poles of the WAC, Central Gondwana and Rio de la Plata APWP's may suggest that its age is closer to 570 Ma. In the absence of other robust paleomagnetic results to constrain the Gondwana APWP between 575 and 525 Ma, we propose a new APWP constrained by the poles B1 and B2 between 575 and 525 Ma for the West and Central Gondwana (Fig. 14d), characterized by a large polar motion between 575 and 565 Ma. Before 575 Ma, the pole WA11 is close to the 525 Ma poles of Central and Eastern Gondwana and Rio de la Plata. It may correspond to a remagnetization of this age as there is no paleomagnetic tests proving its primary nature. It may also define another large polar motion from 615 to 575 Ma in the APWP of the WAC. In that case, the APWP's of Rio de la Plata, Central Gondwana and WAC converge at around 575 Ma which is consistent with the age of the Transbrasiliano suture occurring between 620 and 580 Ma (Caby, 2003; Cordani et al., 2013).

5.3. Global implications

We identified a large polar motion during the Ediacaran in the APWP of the WAC. Let's analyze the APWP's of Laurentia and Baltica.

5.3.1. Apparent polar wander path of Laurentia

In this study, we started from the pole compilation of McCausland et al. (2007) and added the recent results from

Table 3
Paleomagnetic poles from the Gondwana, Laurentia and Baltica for the Ediacaran.

	Rock Unit	Lith.	Unit age (Ma)	Method	Pole (°N)	Pole (°E)	A_{95} or dp/dm	Selection criteria	References
	West African Craton (WAC)								
WA1	Sidi-Said-Maachou Volcanics	Extr	501–513	Strati	18	235	3/6	0-1-1-1-1-1-0	Khattach et al. (1995)
WA2	Sidi-Said-Maachou Volcanics	Extr	501–513	Strati	14	231	4/7	0-1-1-1-1-1-0	Khattach et al. (1995)
WA3	Atar Cliff Group, UnitsCO7&8	Sedi	Cambrian	Strati	0.3	69.3	5.7/11.5	0-1-1-1-0-1-1	Perrin et al. (1988)
WA4	Serie Lie de Vin	Sedi	518–524[a]	Strati	29.2	243.3	10.3/20.5	0-1-0-0-1-1-0	Martin et al. (1978)
WA5	Taroudannt group, volc. & sed.	E&S	551–520[b]	Strati, U-Pb	46.9	42.3	9.7	0-1-1-0-1-1-1	Daly and Pozzi (1977)
WA6	Ait Abdallah 2–4 HT	Sedi	551–520	Strati	−3.7	77.7	8.2/16.5	0-1-0-1-1-1-1	Boudzoumou et al. (2012)
WA7	Ida Ougnidif	Sedi	551–520	Strati	3.4	67.3	8.4/15.6	0-1-0-1-1-1-1	Boudzoumou et al. (2012)
WA8	Abdallah Ida Ougnidif	Sedi	551–520	Strati	−28	54.5	5.7/10.9	0-1-1-1-1-1-0	Boudzoumou et al. (2012)
WA9	Djebel Boho Volc.	Extr	547–526[c]	U-Pb	31	57	17	0-1-1-0-1-1-1	Hailwood (1972)
B2	**Djebel Boho Volc.**	**Extr**	**547–526**	**U-Pb**	**27.3**	**27.1**	**14.9**	**1-1-1-1-1-1-1**	***This study***
A	Ouarzazate Volc.	Extr	577–551	U-Pb	−29.3	56.6	4.1	0-1-1-1-0-1-0	*This study*
B1	**Fajjoud and Tadoughast Volc.**	**Extr**	**572–551**	**U-Pb**	**21.9**	**31**	**15.6**	**1-1-1-1-1-1-1**	***This study***
C	**Adrar-n-takoucht Volc.**	**Extr**	**577–564**	**U-Pb**	**−57.6**	**295.6**	**15.7**	**1-1-1-1-1-1-1**	***This study***
WA10	Tabankort rhyolite	Extr	~590	U-Pb	30	322	10	1-1-0-0-1-0-0	Morel (1981)
WA11	**Adma diorite**	**Intr**	**613 ± 3**	**U-Pb**	**33.5**	**343.5**	**16.1/17.1**	**1-1-1-0-1-0-0**	Morel (1981)
WA12	Nimba supergroup	Meta	620+/50	Rb-Sr	28.1	4.7	40	0-0-1-1-1-1-0	Onstott and Dorbor (1987)
	Amazon craton								
Am1	**Puga Carbonate A**	**Sedi**	**635**	**Strati**	**83**	**34**	**9**	**1-1-1-1-1-1-0**	Trindade et al. (2003)
Am2	Puga Carbonate B	Sedi	525	APWP	34	327	10	0-1-1-1-0-1-0	Trindade et al. (2003)
	Gondwana from Trindade et al. (2006), Rapalini et al. (2015)								
	Central Gondwana								
C9	**Juiz de Fora Complex**	**Meta**	**517.5 ± 17.5**	**Thermo.**	**4.2**	**320.1**	**10**	**1-1-1-1-1-1-1**	D'Agrella-Filho et al. (2004)
C8	Bambui B	Sedi	520	APWP	15	331	3	0-1-1-1-1-1-0	D'Agrella-Filho et al. (2000)
C7	Bambui + Salitre, C	Sedi	520	APWP	32	322	3	0-1-1-1-1-1-0	Trindade et al. (2004)
C6	**Madagascar virgation zone**	**Meta**	**521 ± 12**	**U-Pb**	**−7**	**353**	**14**	**1-1-1-1-1-1-1**	Meert et al. (2003)
C5	**Ntonya Ring**	**Intr**	**522 ± 13**	**Ar-Ar**	**28**	**345**	**1.4/2.3**	**1-1-1-0-1-1-1**	Briden, 1968
C4	**Itabaiana dikes**	**Intr**	**525 ± 5**	**Ar-Ar**	**−34.9**	**134.6**	**8**	**1-1-1-1-1-1-1**	Trindade et al. (2006)
C3	**Sinyai Metadolerite**	**Meta**	**>547 ± 4**	**Ar-Ar**	**29**	**139**	**3/5**	**0-1-1-1-1-1-1**	Meert and Van der Voo (1996)
C2	Mirbat Sediments	Sedi	~700	Strati	23.3	141.8	3.9/7.5	0-1-0-1-1-1-0	Kempf et al. (2000)
NO	**Nola gp. Metadolerite**	**Meta**	**571 ± 6**	**Ar-Ar**	**−61.8**	**304.8**	**5/10**	**1-1-1-1-1-1-1**	Moloto-A-Kenguemba et al. (2008)
C1	**Dokhan Fm. Extrusives**	**Extr**	**593 ± 13**	**U-Pb**	**−43**	**36**	**7/11**	**1-1-1-1-1-1-1**	Nairn et al. (1987)
	Rio de la Plata								
R1	Sierra de las Animas	E&I	520	APWP	5.9	338.1	20/27	0-0-1-1-1-1-0	Sánchez-Bettucci and Rapalini (2002)
Sbc	Cerro Negro fm.	Sedi	520	APWP	11	323.5	10/13	0-1-0-1-1-1-1	Rapalini et al. (2013)
P	Polanco fm.	Sedi	520	APWP	3.2	325.8	15.2	0-1-0-1-1-1-0	Rapalini and Sánchez-Bettucci, 2008
Sbd	Olavarria fm. Sediments	Sedi	535	APWP	21.6	289.9	6/9	0-1-0-1-1-1-1	Rapalini et al. (2013)
LB	Los Barrientos fm. Sediments	Sedi	570	APWP	−15.1	252.6	11/14	0-1-1-1-1-1-1	Rapalini, 2006
R2	**Sierra de las Animas**	**E&I**	**578 ± 4**	**U-Pb**	**−12.2**	**258.9**	**14.9**	**1-1-1-1-1-1-1**	Rapalini et al. (2015)
Sbe	Cerro Largo fm. Sediments	Sedi	580	APWP	−24.8	217.3	8/14	0-1-0-1-1-1-1	Rapalini et al. (2013)
Sbf	Villa Monica fm. Sediments	Sedi	590	APWP	−48.8	198.1	6/11	0-1-0-1-1-1-1	Rapalini et al. (2013)
PH	Playa Hermosa fm. Sediments	Sedi	586 ± 8	Strati, U-Pb	−58.8	183.1	12.1	1-1-0-1-1-1-1	Rapalini et al. (2015)
R1	**Campo-Alegre Fm. Extrusives**	**Extr**	**595 ± 5**	**U-Pb**	**−57.0**	**223.0**	**9**	**1-1-1-1-1-0-1**	D'Agrella-Filho and Pacca (1988)
	Eastern Gondwana (+poles from Mitchell et al., 2010 and Schmidt, 2014*)*								
A2	Todd River Dolomite	Sedi	526–519	fossils	−43.2	339.9	4.5/7.7	1-1-1-1-1-0-1	Kirschvink (1978)
A1	Upper Arumbera Sandstone	Sedi	542–526	Strati	−46.6	337.3	2.6/4.6	1-1-1-1-1-0-1	Kirschvink (1978)
AECr	Upper Arumbera Sandstone	Sedi	542–526	Strati	−59.9	324.3	12.7	1-1-1-1-1-0-1	Mitchell et al. (2010)
A3	**Hawker Group sediments**	**Sedi**	**542–528**	**fossils**	**−21.3**	**14.9**	**6.8/12.5**	**1-1-1-1-1-1-1**	Klootwijk (1980)
ARL	Lower Arumbera Sandstone	Sedi	580–542	Strati	−53.9	348.1	8.8	1-1-1-1-1-0-1	Mitchell et al. (2010)
WK	**Wonoka fm.**	**Sedi**	**575–555**	**fossils**	**−5.2**	**30.5**	**3.6/6.8**	**1-1-1-1-1-1-1**	Schmidt and Williams (2010)
BU	Bunyeroo fm.	Sedi	593 ± 32	Rb-Sr	−18.1	16.3	6.5/11.8	0-1-1-1-1-1-1	Schmidt and Williams (1996)
BF	Brachina fm.	Sedi	609 ± 64	Rb-Sr	−46.0	315.4	2.4/4.6	0-1-1-1-1-1-1	Schmidt and Williams (2010)
	Laurentia from McCausland et al. (2007, 2011) and Halls et al. (2015)								
SC	**Skinner Cove Volcanics**	**Extr**	**550 ± 3**	**U-Pb**	**−16**	**338**	**9**	**1-1-1-1-1-0-1**	McCausland and Hodych (1998)
JR	Johnnie Fm. (unrotated)	Sedi	635–550	Strati	−10	162	5/10	0-1-1-1-0-1-0	Van Alstine and Gillett (1979)
SIa	**Sept-Iles intrusion (shallow)**	**Intr**	**565 ± 4**	**U-Pb**	**−20**	**321**	**5/9**	**1-1-1-1-1-1-1**	Tanczyk and Lapointe (1987)
SIb	Sept-Iles intrusion (steep)	Intr	565 ± 4	U-Pb	61	295	10	0-1-1-1-1-1-1	Tanczyk and Lapointe (1987)
CTb	Catoctin Basalts (shallow)	Extr	572 ± 5	U-Pb	4	3	7/13	0-1-1-1-1-1-0	Meert et al. (1994)
CTa	**Catoctin Basalts (steep)**	**Extr**	**572 ± 5**	**U-Pb**	**42**	**297**	**17**	**1-1-1-1-1-1-1**	Meert et al. (1994)
BF	Buckingham flows (in-situ)	Extr	573 ± 32	K-Ar	0	351	5/9	0-1-1-1-1-1-1	Danker and Lapointe (1981)
CC	**Callander Alkaline Complex**	**Intr**	**577 ± 1**	**U-Pb**	**46**	**301**	**6**	**1-1-1-1-1-1-1**	Symons and Chiasson (1991)
BMa	**Baie des moutons Complex**	**Intr**	**583 ± 2**	**Ar-Ar**	**42.6**	**332.7**	**11.7/12.4**	**1-1-1-1-1-1-1**	McCausland et al. (2011)
BMb	Baie des moutons Complex	Intr	583 ± 2	Ar-Ar	−34.2	321.5	10.9/21.8	0-1-1-1-1-1-0	McCausland et al. (2011)
GD1	Grenville dikes (shallow)	Intr	590 ± 2	U-Pb	3	331	10/15	0-1-1-1-1-1-0	Murthy (1971)
GD2	**Grenville dikes (steep)**	**Intr**	**590 ± 2**	**U-Pb**	**60**	**240**	**14/14**	**1-1-1-1-1-1-1**	Murthy (1971)

(*continued on next page*)

Table 3 (*continued*)

	Rock Unit	Lith.	Unit age (Ma)	Method	Pole (°N)	Pole (°E)	A_{95} or dp/dm	Selection criteria	References
GDB	**Grenville dikes (steep)**	**Intr**	**587.3 ± 0.7**	**U-Pb**	**233.4**	**55.7**	**10.6**	**1-1-1-1-1-1-1**	Halls et al. (2015)
GDC	Grenville dikes (shallow)	Intr	584.8 ± 0.6	U-Pb	331.3	−20.3	8.6	0-1-1-1-1-1-0	Halls et al. (2015)
GDD	Grenville dikes (shallow)	Intr	585.2 ± 0.8	U-Pb	338.4	0.1	13.3	0-1-0-1-1-1-0	Halls et al. (2015)
GDE	**Grenville dikes (steep)**	**Intr**	**586 ± 4**	**U-Pb**	**231.2**	**51.5**	**10.9/12.3**	**1-1-1-1-1-1-1**	Hyodo and Dunlop (1993)
LR	**Long Range dikes**	**Intr**	**615 ± 2**	**U-Pb**	**19**	**355**	**18/25**	**1-0-1-1-1-1-1**	Murthy et al. (1992)
	Baltica from Klein et al. (2015)								
TK	Tornetrask formation	Sedi	~535	fossils	56	116	13.4	0-1-1-1-1-1-0	Torsvik and Rehnström (2001)
NS	Nekso sandstone	Sedi	~545	Strati	40	170	5.1	0-1-0-1-1-1-0	Lewandowski and Abrahamsen (2003)
ZN	**Zigan formation clastic rocks-C**	**Sedi**	**547.6+/3.8**	**U-Pb**	**16.2**	**318.4**	**4.1**	**1-1-1-1-1-1-1**	Levashova et al. (2013)
V	**Verkhotina sediments, comp. Z**	**Sedi**	**550 ± 5**	**U-Pb**	**31.7**	**292.9**	**2.1**	**1-1-1-1-1-1-1**	Popov et al. (2005)
ZA	**Zolotitca sediments, comp. B**	**Sedi**	**550 ± 5**	**U-Pb**	**28.3**	**290**	**3.3**	**1-1-1-1-0-1-1**	Llanos et al. (2005)
UA	Volhyn lavas – combined	Extr	~580	Ar-Ar	20.0	4.4	28.3	1-0-1-1-1-1-1	Elming et al. (2007)
WC	**Winter coast sediments, com. Z**	**Sedi**	**555 ± 3**	**U-Pb**	**25.3**	**312.2**	**2.9**	**1-1-1-1-1-1-1**	Popov et al. (2002)
Ch	**Chernokamenskay group C**	**Sedi**	**557 ± 13**	**U-Pb**	**17.3**	**306.7**	**6.0**	**1-1-1-1-1-1-1**	Fedorova et al. (2014)
Po	**Podolia, comp. V**	**Sedi**	**545–570**	**Strati**	**40**	**276.5**	**7.5**	**1-1-1-1-1-1-1**	Iosifidi et al. (2005)
BA	**Basu formation**	**Sedi**	**570–560**	**Strati,fossils**	**−2**	**6.0**	**3.8**	**1-1-1-1-1-1-1**	Levashova et al. (2015)
KL	Krivaya Luka formation	Sedi	600–570	APWP	58.6	275.8	6.1	0-1-1-1-1-1-1	Lubnina et al. (2014)
KF	**Kurgashlya formation**	**Sedi**	**600–570**	**Strati**	**50.9**	**314.5**	**5.3**	**1-1-1-1-1-1-1**	Lubnina et al. (2014)
BK	**Bakeevo formation**	**Sedi**	**600–570**	**Strati**	**42.3**	**299.1**	**5.3**	**1-1-1-1-1-1-1**	Lubnina et al. (2014)
F	Fen Complex	Intr	583 ± 15	Ar-Ar	53.2	145.7	5.6	1-1-1-1-1-1-0	Meert et al. (2007)
Ast	Alnö carbonate complex (steep)	Intr	584 ± 7	Ar-Ar	62.7	101.2	13.3	1-1-1-1-1-1-0	Meert et al. (2007)
Ash	Alnö carbonate complex (shallow)	Intr	584 ± 7	Ar-Ar	3.5	269	23.2	1-0-1-1-0-1-1	Meert et al. (2007)
H	Hailuoto sediments	Sedi	660–550	Strati,fossils	48.7	241.1	7.6	0-1-1-1-1-1-1	Klein et al. (2015)
ED	**Egersund dykes**	**Intr**	**616 ± 3**	**U-Pb**	**31.4**	**44.1**	**15.6**	**1-1-1-1-1-1-1**	Walderhaug et al. (2007)

Selection criteria (1–7) are summarized as follows: (1) unit age error <20 Ma and presumption that the magnetization age is similar to the unit age, (2) statistical precision (A_{95} <16°), (3) adequate sample number (Number of sites >4 and number of samples >24), (4) detailed demagnetization illustrated via vector subtraction or principal component analysis (Kirschvink, 1980), (5) no negative paleomagnetic tests, (6) structural control/continuity with the craton, (7) dissimilarity from more recent paleomagnetic directions except if a paleomagnetic test demonstrates that a remagnetization of this age is not possible.

[a] Age following Maloof et al. (2010).

[b] Age following Gasquet et al. (2005).

[c] Age following Ducrot and Lancelot (1977), Gasquet et al. (2005), Blein et al. (2014a).

McCausland et al. (2011) and Halls et al. (2015). The poles are distributed in two groups at high and low latitude respectively (Fig. 15). The APWP constructed with these poles displays large oscillations between 615 and 550 Ma, as already suggested in the literature (Abrajevitch and Van der Voo, 2010). Most of the large oscillations rely on the identification of two components in a same study (Murthy, 1971; Tanczyk and Lapointe, 1987; Meert et al., 1994; McCausland et al., 2011; Halls et al., 2015). McCausland et al. (2007) discussed the robustness of each pair of directions and argued that there is always a pole more reliable than the other. For instance, the steep component B of the 565 Ma Sept-Iles intrusion is carried by multi-domain magnetite and is probably not a primary magnetization. The recent work of Bono and Tarduno (2015) supports this view, showing that the component B is a viscous remagnetization. On the contrary, the component A is most probably primary since it corresponds to a thermochemical remanence carried by single-domain magnetite and is supported by a positive reversal test. The component A of the Catoctin basalts (Meert et al., 1994) is the only one supported by a baked contact test while the component B may represent a prefolding Cambrian overprint (Meert et al., 1994; McCausland et al., 2007). For the Grenville dyke swarm (Murthy, 1971), McCausland et al. (2007) consider that only the steep component (GD2) is a primary component. The shallow one (GD1) is not supported by any field test and may correspond to a Cambrian remagnetization. Halls et al. (2015) recently conducted an extensive paleomagnetic study in the Grenville dyke swarm in order to clarify previous paleomagnetic studies (e.g. Murthy, 1971; Hyodo and Dunlop, 1993). They observed a complex pattern displaying a large swath of paleomagnetic directions along a great circle of NW-SE trend. Halls et al. (2015) argue that these components are primary and propose that several Grenville dykes record an equatorial dipole field as a transitional field during a highly unstable regime of the geodynamo. More specifically, four components were interpreted as primary. The components B and E (referring to the poles GDB and GDE in Fig. 15) are steep and antipodal, and probably refer to the same paleomagnetic component. We notice that the component E from Mattawa dykes has been redated by Halls at 585.9 Ma, close to their 587.3 ± .7 datation of Key River dykes (component B). The same component, called GD2, was already observed by Murthy (1971) and Hyodo and Dunlop (1993), and its primary nature is supported by a positive reversal and contact test (Hyodo and Dunlop, 1993; Halls et al., 2015). Moreover, this component is observed in a same dyke over a long distance of some 150 km (Halls et al., 2015). This component is most probably a primary component. The components C and D (referring to the poles GDC and GDD in Fig. 15) are shallow directions almost antipodal and may refer to the same paleomagnetic component (GDC and GDD; Fig. 15). These components gave inconclusive contact tests and one sample carries both this component and the primary component B. In spite of its continuity along a same dyke, we cannot exclude that this component is a remagnetization. If so, the directions B and E are the only primary components recorded in the Grenville dyke swarm, suggesting that a much more stable magnetic field than proposed by Halls et al. (2015) existed at that time. McCausland et al. (2011) observed a similar pair of poles in the Baie des Moutons syenite (BMa and BMb). Both poles may correspond to primary components even if there is no paleomagnetic tests to support their primary nature. BMa is close to other high latitude poles of similar age and may refer to the primary component whereas BMb, geologically younger than BMa, is closer to younger Ediacaran poles. Its location, close to the poles C and D from Halls et al. (2015) is close to the Sept Iles A pole (which appears to be a robust key pole as discussed above), and may indeed indicate a remagnetization of this age. This is our preferred scenario while alternative explanation such as the one proposed by Halls et al. (2015), the record of

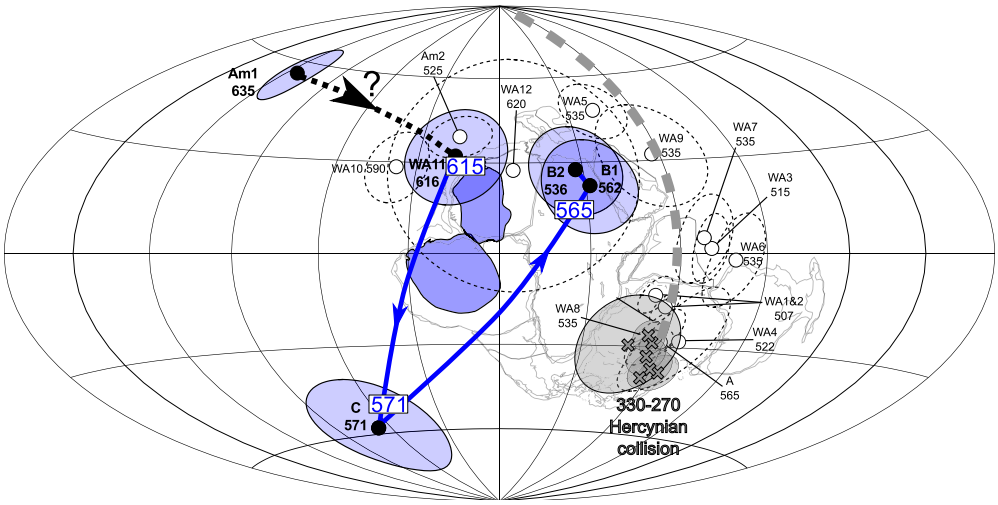


Fig. 13. Poles from the West African craton (WAC) and Amazon craton determined in this study and compiled from the literature. The abbreviations refer to the poles presented in the Table 3. The mean age of each pole is indicated in Ma. The poles with their 95% ellipse of confidence colored in blue are selected in this study whereas the others are rejected. The 330–270 Ma segment of the APWP of Gondwana (running mean type; Torsvik et al., 2012) is represented by grey crosses. The grey broken great circle segment connects the position of Carboniferous-Permian remagnetization and the present-day field. Poles lying along the arc may reflect a mixture of the modern field with a late Paleozoic overprint. The contours of the WAC and the Amazon craton are filled in blue. The contour of the Gondwana is represented in light grey. The projection is Hammer-Aitoff. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

an equatorial dipole field as a transitional field between opposite polarity states cannot be completely ruled out. Two other poles are also used in the compilation of McCausland et al. (2007): the 550 Ma Skinner Cove pole and the 615 Ma Long Range dyke pole. The first result comes from the Western Newfoundland, which is allochtonous to Laurentia. Hodych et al. (2004) argue that this pole was never far from Laurentia and may be used in the construction of the Laurentia APWP. The Long Range dyke pole is supported by a positive contact test, is observed in reverse and normal polarities and is the only one defining the position of Laurentia for early Ediacaran times. In spite of its poorly spatial definition (dp = 18°, dm = 25°), we will use it in the construction of the APWP of Laurentia. Finally, the Johnnie formation (JR) and the Buckingham flows (BF) poles do not fullfill our age criteria and were not retained in our selection. To sum up, the APWP proposed by McCausland et al. (2007) seem to be a robust solution, displaying a single oscillation from low latitude at 615 Ma to high-latitude around 590–575 Ma and then back to low latitude at 565 Ma (Fig. 15).

5.3.2. Apparent polar wander path of Baltica

Large oscillations in the APWP of Baltica are observed during the Ediacaran (Abrajevitch and Van der Voo, 2010), as it was the case for Laurentia. In this study, our starting point was the compilation of Klein et al. (2015) which integrates the newest results (Fig. 16). The large oscillations proposed by Abrajevitch and Van der Voo (2010) are constrained by the poles TK, F and Ast, from respectively the Torntrask formation (Torsvik and Rehnström, 2001), the Fen complex (Meert et al., 2007), and the Alnö carbonate complex (Meert et al., 2007). All three results display pole directions close to those of the Permo-Trias APWP of Baltica, none of them supported by paleomagnetic tests. Therefore, they likely represent remagnetizations acquired during this period of time, that may correspond to the opening of the Oslo Rift, as already suggested for the Fen Complex result by Meert (2014). The pole KL is a remagnetization dated only by the comparison with other paleomagnetic poles and is not selected either. The poles UA and Ash have large ellipses of confidence at 95% level, respectively of 28.3° and 23.2° and were not retained either. Finally, the pole H from the Hailuoto sediments was not selected because it is illdated and does not pass our age criteria. The APWP defined by the selected poles (Table 3) displays back and forth movements from 616 Ma to 550 Ma. This APWP is similar to the one proposed by Klein et al. (2015) except around 575 and 555 Ma where we included the 565 Ma result from the Basu formation (BA). This result is close to the 440 Ma segment of the Baltica APWP (Torsvik et al., 2012) but the positive reversal test, fold-test and slump tests support a primary origin. Meert (2014) argues that this pole is derived from the margin of Baltica, and may suffer local vertical axis rotations. Nevertheless, there is no trace of collisional events from the late Neoproterozoic up to the Permian (Puchkov, 2003; Levashova et al., 2015). Moreover, the pole ZN determined in the same region than the pole BA is close to coeval poles identified in the craton (Ch, WC, ZA, V) which support the absence of large relative motion between the studied area and the craton (Levashova et al., 2013). Levashova et al. (2013) also observed a Permian synfolding remagnetization in the Zigan formation rocks which is consistent with the stable Europe APWP, ruling out a large rotation of this formation during the Permian and later compared to the craton. We are aware that the loop is supported by a single data whose position had remained unexplained until now, but this pole passes all our selection criteria and does not seem to have been affected by strong rotation around a vertical axis.

5.3.3. Is the loop compatible with Ediacarian paleogeographic reconstructions and the Iapetus rifting system?

We identified large polar motions between 615 and 565 Ma in the APWP's of the WAC, Laurentia and Baltica. We have performed a fit in Fig. 17a by superimposing the 575–565 Ma segments of WAC, Laurentia and Baltica. Each path shows consistent amplitudes and shapes where poles dated respectively around 615, 590–575 and 565 Ma formed three distinct groups. Inside each group, the 95% ellipses of confidence of the poles intersect. The Euler poles of rotation derived from this fit were used to reconstruct the paleogeography at 575 Ma (Fig 17b). Laurentia lay in the vicinity of the south pole whereas the West and Central Gondwana and Baltica lay around tropical regions (Fig. 17b). The relative position of these three large continents is consistent with the Iapetus opening system (e.g. Cawood et al., 2001; Cawood and Pisarevsky, 2006; Pisarevsky et al., 2008): (1) the eastern margin of Laurentia faces to the western margin of the West Gondwana

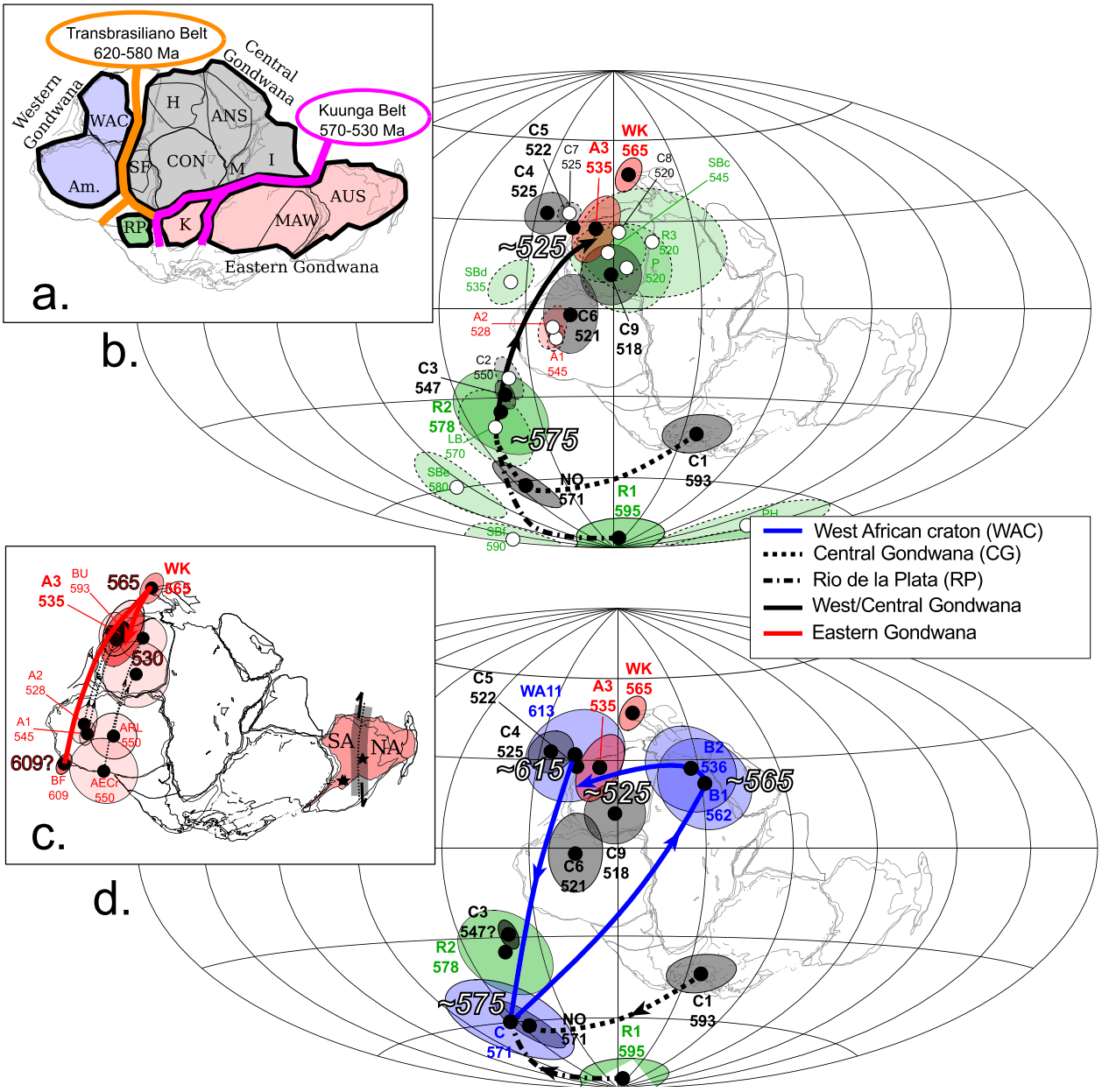


Fig. 14. a. Simplified map showing the position and age of the main sutures involved in the accretion of the Gondwana. Abbreviations refer to the cratons: WAC: West Africa, Am.: Amazonia, H: Hoggar, SF: Sao Francisco, CON: Congo, K: Kalahari, I: India, MAW: Mawson, AUS: Australia b. APWP of the Gondwana using the poles compiled by Trindade et al. (2006) and Rapalini et al. (2015). Poles used in its construction are represented, and their abbreviations presented in Table 3. The color of the 95% ellipse of confidence indicates the block from which it is representative. Black (white) points refer to poles (not) passing our selection criteria. c. Simplified map showing the large dextral movement associated to the Paterson and Petermann orogens between the North Australian craton (NA) and the South Australian cratons (SA). Poles from the compilation of Trindade et al. (2006), Mitchell et al. (2010) and Schmidt (2014) are represented, and their abbreviations presented in Table 3. Dark (light) 95% ellipse of confidence refer to poles from SA (NA).The red path refer to the APWP of Australia (Schmidt, 2014) constructed with poles from SA only. When corrected from the relative movement between NA and SA (Euler pole from Li and Evans, 2011), poles from NA move close to the APWP of Schmidt (2014). d. Proposition of a new APWP based on the poles from this study in blue. Euler parameters for the Gondwana fit relative to Central Africa: South America: 50.0°N, −32.5°E, 55.52° (Seton et al., 2012), West Africa: 33.6°N, 26.0°E, 2.3° (Torsvik et al., 2012), Madagascar: −14.8°N, 317.5°E, 15.3° (Müller et al., 2008), Australia: −19.6°N, −62.2°E, 56.2° (Müller et al., 2008).

consistent with the opening of the western Iapetus ocean; (2) the north-eastern margin of Laurentia faces to the western margin of Baltica consistent with the opening of the eastern Iapetus ocean; (3) the southern margin of Baltica faced to the north-western margin of the West Gondwana consistent with the opening of the Tornquist ocean. The analysis of the age of the rift-related mafic magmatism on each of these margins places the opening between 615 and 530 Ma (Pisarevsky et al., 2008) compatible at least at first order with these geological observations. Consequently, the large polar motions observed in the APWP's of WAC, Laurentia and Baltica between 615 and 565 Ma do not generate large relative movements between these blocks and is roughly in agreement with a geocentric dipole field during the Ediacaran. We also remark the existence of antipodal normal and reverse polarity directions for several of the poles for this period of time (Abrajevitch and Van der Voo, 2010), whatever they are equatorial or polar, and seems to rule out the existence among these data of remagnetizations or unseparated magnetic components as suggested for example by Hodych et al. (2004).

In the APWP of Baltica, a large polar wander is also found between 565 and 550 Ma (Fig. 16). This feature is not observed in the reappraised APWP's of Gondwana and Laurentia (Figs. 14d

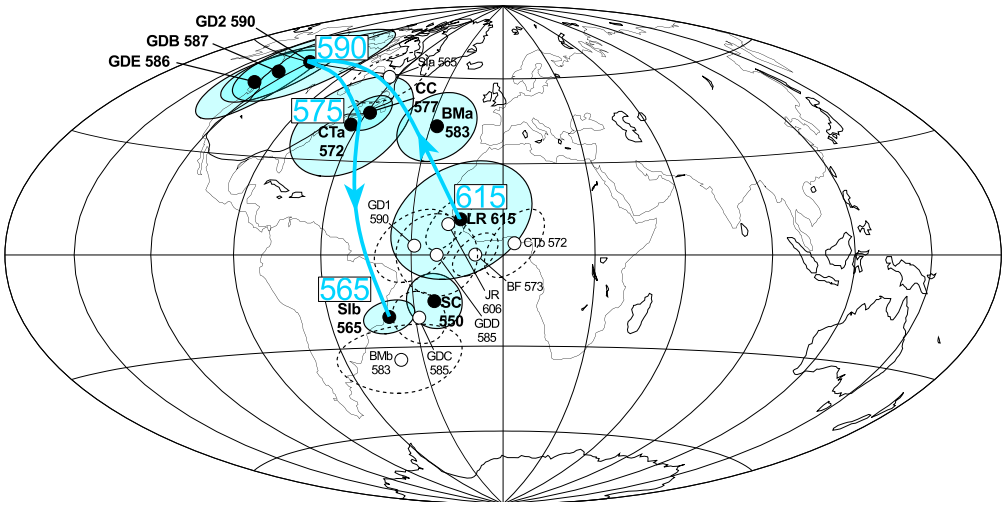


Fig. 15. Poles from Laurentia following the compilation of McCausland et al. (2007) and new results from McCausland et al. (2011) and Halls et al. (2015). Black (white) poles (do not) pass our selection criteria. The cyan line represents the APWP of Laurentia constructed based on the cyan poles. The abbreviations refer to the poles presented in Table 3. The mean age of each pole is indicated in Ma. The contours of Laurentia and the continents in present-day coordinates are represented in the background. The projection is Hammer-Aitoff.

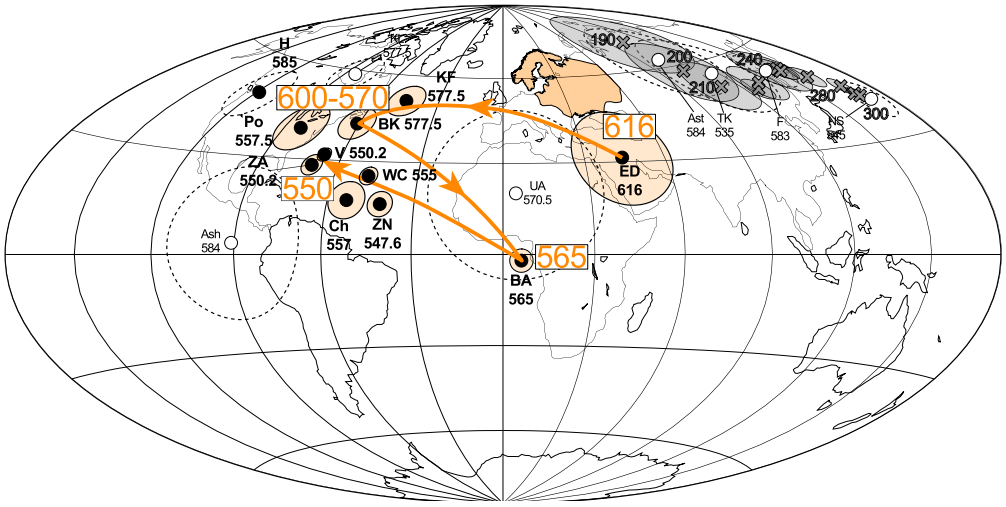


Fig. 16. Poles from Baltica following the compilation of Klein et al. (2015). Black (white) poles (do not) pass our selection criteria. The orange line represents the APWP of Baltica constructed based on the selected poles. The abbreviations refer to the poles presented in Table 3. The mean age of each pole is indicated in Ma. The 190–300 Ma segment of the APWP of Baltica (Running mean type; Torsvik et al., 2012) is represented by grey crosses. The contours of Baltica and the continents in present-day coordinates are represented in the background. The projection is Hammer-Aitoff.

and 15) and does not seem to represent a global feature such as TPW or magnetic field perturbation. This record could be the signature of plate motion, involving a large rotation of Baltica for the late Ediacaran.

5.4. Geodynamic implications

Our paleomagnetic study shows the existence of two large polar events at the same period as those observed in Baltica and Laurentia. The good agreement in shape and time of these large polar motions suggests that the presence of artifacts in these data is unlikely. Three other hypothesis can explain these APWP's: large episodes of True Polar Wander, oscillations of an Earth's magnetic field governed by an axial and an equatorial dipole or the record of an equatorial dipole field as a transitional field between opposite polarity states. In the following discussion, we call event I the large polar motion occurring between 615 and 590–575 Ma, and event II between 575 and 565 Ma. The velocity of these polar motions is critical to assess the responsible mechanism. We calculated the average velocity of the APWP's of WAC, Baltica and Laurentia by using each of their constitutive poles. The minimum and maximum velocities were calculated using the A_{95} and age error of each pair of poles. For the event I (Fig. 18a), the average velocities for the WAC, Baltica and Laurentia are comprised between 2 and 4°/Ma. These velocities are probably underestimated as this large polar motion could have finished earlier than 590/575 Ma, but the lack of poles dated between 615 and 590/575 Ma rules out a more accurate estimation. In the APWP of Laurentia, the event I ends as early as 590 Ma, defined by the Grenville dykes poles (Murthy, 1971; Halls et al., 2015), and may constrain the upper bound duration of this event at around 25 Ma. The mean velocity for these three APWP's is around 3°/Ma even if a better estimation should be calculated between 615 and 590 Ma, giving 3.6°/Ma. For the event II, the average velocities are comprised between 3 and 18.5°/Ma,

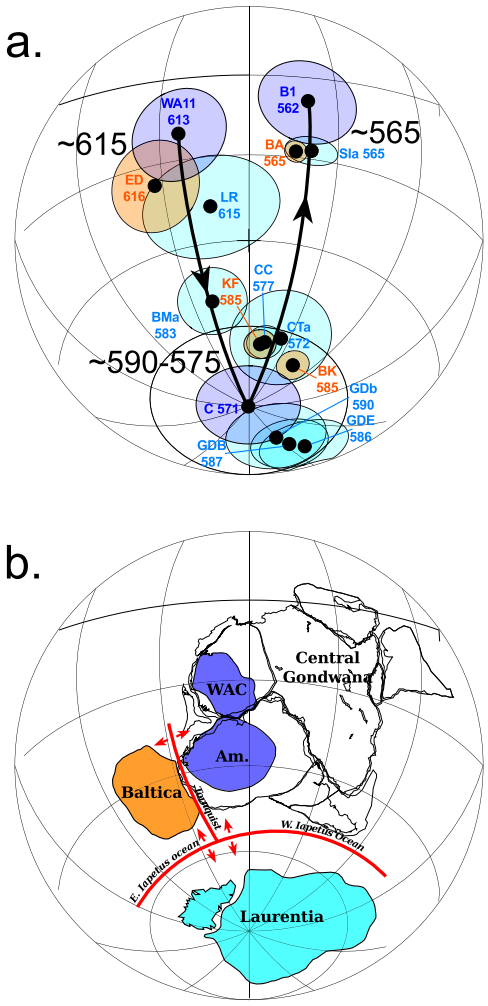


Fig. 17. (a). Fit of the 575–565 Ma segments of the APWP's of WAC, Laurentia and Baltica. The selected poles (mean age in Ma as shown) from each continent is distinguished by the color of the 95% ellipse of confidence: dark blue for WAC, light blue for Laurentia and orange for Baltica. The pole C from this study is used as the reference pole for the reconstruction at ~575 Ma. b. Corresponding paleogeography at ~575 Ma. The color of the blocks refer to the color of the poles described in the figure a. Red lines correspond to ridges associated to the opening of the eastern (E.), western (W.) Iapetus oceans and the Tornquist ocean. Euler parameters used: West African craton: −3.4°N, 207.0°E, −32.4°, Baltica: −1.6°N, 190.2°E, −145.3°, Laurentia: −2.8°N, 349.2°E, 156.1°.

giving a mean velocity for the three APWP's of 10°/Ma. For the WAC, this large polar motion occurred between the emplacement of the Adrar-n-Takoucht and Tadoughast formations. For the pole B1, we thus used the mean age and error of the Tadoughast formation instead of using the mean age of the Tadoughast and Fajjoud formations in the calculation of the event II velocity. Notice here that the average velocity for the APWP of the WAC is much higher than those from Laurentia and Baltica. However, its lower-bound velocity is in good agreement with the other mean values which do not exceed 9°/Ma. For the APWP of Baltica, the average velocities were calculated between 585 to 565 Ma. For the WAC and Laurentia, the same event lasts 10 Ma. Consequently, the velocities from the APWP of Baltica are certainly underestimated.

TPW has already been invoked to explain the large polar motions observed in the Ediacaran (Kirschvink et al., 2005; McCausland et al., 2007; Mitchell et al., 2011). TPW is the coherent motion of Earth's lithosphere and mantle taken as a whole with respect to the rotational axis and is due to the angular momentum conservation (Munk and MacDonald, 1960). On geological timescale, the main sources of inertia perturbations are the mass heterogeneities inside the mantle. More specifically, it can be related to the cold slabs sinking into the mantle and the hot materials going up from the base of the mantle. The time-scale upon TPW occurs is dependent on the viscosity of the mantle which controls the readjustment of the equatorial bulge, the amplitudes of the mass heterogeneities inside the mantle and the strength of the lithosphere (Creveling et al., 2012). Tsai and Stevenson (2007) estimated a speed limit for the TPW of around 2.5°/Ma based on the current rheological parameters of the Earth. This estimation is based on the current Earth's inertia, stabilized by the peri-Pacific-like girdle of subduction. Larger rate of TPW have been estimated in cases of larger mass reorganizations such as those associated with the formation of a supercontinent, reaching up to around 10°/Ma (Greff-Lefftz and Besse, 2014). In this geodynamical context, a large pair of 90° TPW events can be generated due to an inertial interchange associated with the formation of a girdle of cold downwellings encircling the supercontinent (Greff-Lefftz and Besse, 2014) or to the rising of a hot mantle plume at the center of the supercontinent (e.g. Zhong et al., 2007). Large oscillating TPW can be also generated if an elastic lithosphere is considered (Creveling et al., 2012). The large oscillation in the APWP's of WAC, Laurentia and Baltica is compatible in velocity and shape to an IITPW. We have also tested the TPW hypothesis using a simple model of mantle dynamics (Robert, 2016) using the same formalism as Greff-Lefftz and Besse (2014). Our results, although preliminary, seem to indicate that the IITPW hypothesis during

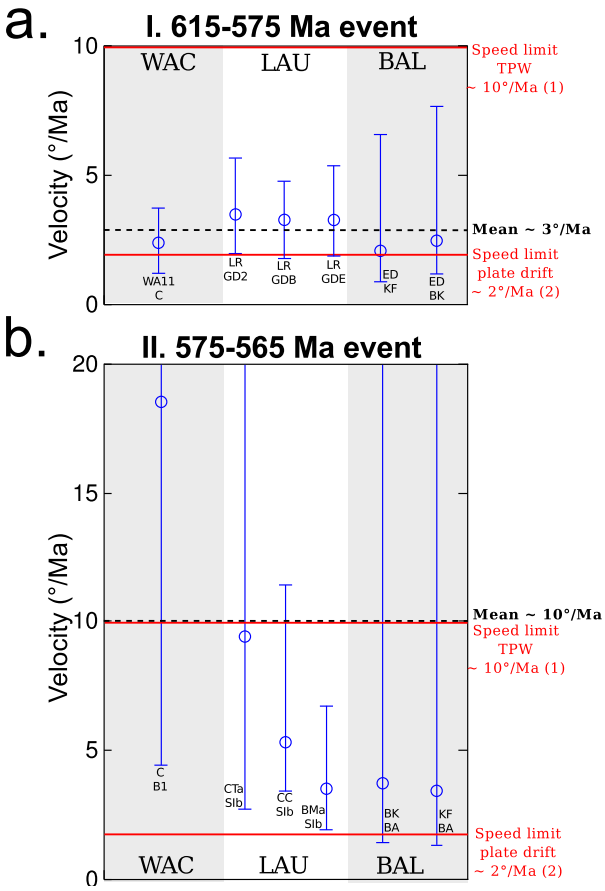


Fig. 18. Velocities of the 615 to 590–575 Ma (a.) and the 575 to 565 Ma (b.) polar motions for each APWP. Abbreviations are: WAC: West African craton, LAU: Laurentia, BAL: Baltica. Velocities are calculated based on each pair of poles displaying these large polar motions. The abbreviated name of each pair is noted close to each point. The pole abbreviations are the same as those defined in table 3. Error bars were calculated using the full range of age and A_{95} errors. The mean is calculated based on the average velocities from each APWP. Speed estimations comes from (1) Greff-Lefftz and Besse (2014) and (2) Zahirovic et al. (2015).

the Ediacarian is possible. The pole drift rate reaches indeed about 6°/Ma for the second episode of TPW. Therefore, if large changes of mass heterogeneities occurred during the Ediacaran, this mechanism could explain the paleomagnetic data for this period of time, as already proposed by McCausland et al. (2007) among others. Both the dislocation of Rodinia and the formation of Gondwana during pre-Ediacaran to Ediacaran times can dramatically change the mass distribution inside the mantle. Other indicators such as the hotspot tracks observed on continents can give insights on the TPW hypothesis. Mitchell et al. (2011) argue that the Sutton hotspot, which is located on the eastern part of Laurentia during the Ediacaran, generate a quasi-static track, indicating that the large paleomagnetic oscillations observed in the APWP of Laurentia is not produced by plate drift but by TPW episodes.

Oscillations of a magnetic field governed by an axial and an equatorial dipole could also explain the paleomagnetic data, as already proposed by Abrajevitch and Van der Voo (2010). Equatorial dipoles have been found in weak-field numerical dynamo regimes for parameters far from those of the Earth (Aubert and Wicht, 2004). However, a bistable regime between an equatorial and axial dipole could be generated for the Earth during a reversal in the case of large heat flux heterogeneities at the core mantle boundary (Gissinger et al., 2012). In this bistable regime, the axial magnetic field exhibits chaotic switches of its polarity, involving a transitional equatorial dipole during frequent polarity reversals. In our study, we were not able to assess the frequency reversal rate as the volcanic rock record is too heterogeneous in time. Nevertheless, our results are not incompatible with very high frequency reversal rates already proposed for the Ediacaran and Cambrian (Halls et al., 2015; Bazhenov et al., 2016; Gallet et al., 2003) if rapid volcanic extrusions are considered. If the components B and C found in this study were produced by the alternation of an axial and an equatorial dipole field, the equatorial dipole field has to always occur at the same longitude (Halls et al., 2015). Kutzner and Christensen (2004) show that a large-scale heterogeneity of the heat flux at the core-mantle boundary could generate a VGP preferred path during reversals. Abrajevitch and Van der Voo (2010) and Halls et al. (2015) speculate that the Sutton plume, located on the eastern side of Laurentia, may have constrained the position of the equatorial dipole over Laurentia. In such a case, the poles C, H, BK, KF, CC, CTa and GDb, located at high latitude over Laurentia in the Fig. 17, could be the record of the equatorial dipole field. In this scenario, the 615 Ma poles and 565 Ma poles shown in Fig. 17 would describe an axial dipole field. If it is true, Laurentia for example would have remained at low latitude during all the Ediacaran whereas the West African craton would have evolved at high latitude only.

6. Conclusions

Large and fast oscillations of the palomagnetic poles in the APWP's of Laurentia and Baltica have led to several hypothesis: (1) unreliable paleomagnetic data, (2) highly unusual equatorial geomagnetic dipole field during this period, (3) recording of an equatorial dipole field as a transitional field between opposite polarity states and (4) inertial interchange true polar wander. In order to test these hypotheses, we conducted a paleomagnetic study on volcanic rocks of Ediacaran and Cambrian age from the Anti-Atlas. We identified a Hercynian remagnetization component (A) and two other components (B and C). The component B is supported by a positive fold-test constraining its age before the Hercynian orogeny and a positive intraformational conglomerate test suggests that it is a primary component. The positive fold-test observed for the component C constrains its age before the Ediacaran fold and supports its primary nature. Based on stringent selection criteria, we have constructed an APWP of the WAC for the Ediacaran showing a large oscillation between 615 and 565 Ma. We have also reappraised the APWP's of Gondwana, Laurentia and Baltica. The APWP of Gondwana is similar to that of WAC at 575 and 525 Ma, but the absence of intermediate data precludes any estimation of the polar motion amplitude. The APWP's of Laurentia and Baltica display a similar large oscillation, consistent in shape, amplitudes and ages. A global reconstruction based onto the APWP's superimposition leads to a paleogeographic solution which is compatible with the opening of the Iapetus and Tornquist oceans during the Ediacaran. Therefore, the hypothesis of paleomagnetic data artifacts creating the large oscillation is now unlikely. Modeling of TPW show that the range of polar wander velocities encountered in the Ediacaran paleomagnetic studies is compatible with an IITPW hypothesis if large perturbations of the mass distribution inside the mantle occurred. Consequently, our preferred explanation to explain our signal goes toward an IITPW process. Large continental reorganizations due to the Rodinia dislocation and the accretion of the Gondwana during the Ediacaran may have played a major role in the Earth inertial evolution. Dramatic climatic changes due to large latitudinal variations are expected in case of TPW, whereas only slight variations in latitude would occur in the field hypothesis. Modelling the Ediacaran climate using our APWP, along with a compilation of paleoclimatic indicators will help to solve this problem in the future, and needs to be considered to bring additional constrain to this scientific question. Finally, the paleomagnetic signal proves to be extraordinary complicated in most studies covering this time and encourages caution. More paleomagnetic and radiometric data are still necessary to ascertain the mechanisms involved in this rapid polar change, including continuous records in order to assess the existence of rapid plate motion or magnetic field perturbations.

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